Towards an Incentive-Compatible Decentralized Ecosystem for the Agent Society

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Contents

1 Introduction **3**
1.1 From Language Models to an Agent Society . . . 3
1.2 The Trust Crisis in Agent Collaboration . . . 3
1.3 An Incentive-Compatible Ecosystem . . . 4
1.4 Contributions . . . 5

2 An Incentive-Compatible Ecosystem for Agent Society **6**
2.1 Agent Roles and Agent Services . . . 6
2.2 Agent Actions and Validation Protocol . . . 7
2.2.1 Promotion and Subscription . . . 8
2.2.2 Validation Requests . . . 9
2.2.3 Consensus Formation and Proof-of-Intelligence . . . 9
2.3 Tokenomics Incentive Structure . . . 11
2.3.1 Staking, Rewards, and Penalties . . . 11
2.3.2 Token Flow and the Tokenomics–Knowledge Dual Loop . . . 12
2.3.3 Protocol Summary . . . 14

3 Theoretical Foundations and Empirical Validation **14**
3.1 Game-Theoretic Guarantees . . . 14
3.1.1 Model Setup and Assumptions . . . 15
3.1.2 Incentive Compatibility of Validators . . . 16
3.1.3 Producer's Threshold-Optimal Promotion . . . 17
3.1.4 Consumer's Threshold-Optimal Subscription . . . 17
3.1.5 Symmetric Bayesian–Nash Equilibrium . . . 18
3.1.6 Robustness: Relaxing the Idealized Assumptions . . . 19
3.2 Case Study: Decentralized Question Answering (DeQA) . . . 21
3.2.1 Instantiation . . . 21
3.2.2 Simulation Setting . . . 22

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3.2.3 Agent Rationality and Community Evolution (RQ1) 23
3.2.4 Validation Mechanism Analysis (RQ2) 23
3.2.5 Service Quality Impacts (RQ3) 24
3.3 Application Scope 24

4 Related Work **25**

5 Limitations and Future Work **26**
5.1 Limitations 26
5.2 Future Work 26

6 Conclusion **27**

A Proofs and Technical Details **32**

Abstract

As Large Language Models (LLMs) evolve into autonomous agents, the emerging landscape signifies a fundamental transition toward an open agent society—a decentralized population of specialized actor agents that discover, evaluate, and transact across organizational and domain boundaries. Since no single agent can achieve universal mastery, sustainable agent collaboration mandates a web of cross-organizational dependencies among self-interested actors who must be properly incentivized to participate both diligently and honestly. However, unlike existing multi-agent systems that rely on a centralized orchestrator, our envisioned decentralized ecosystem admits no such authority and remains structurally vulnerable to a foundational trust crisis: the simultaneous collapse of cooperation, accountability, and verifiability triggers an adverse selection spiral that threatens to unravel the agent society into a "market for lemons."

To resolve trust crisis, we envision a decentralized ecosystem where autonomous agents coordinate through an incentive-compatible mechanism. This ecosystem organizes agents into dynamically interchangeable roles: consumers who issue token-backed service requests, producers who promote and provide specialized services, and validators who apply domain intelligence to assess service quality. To facilitate high-stakes transactions under information asymmetry, we introduce Proof-of-Intelligence (PoI), a decentralized validation paradigm that incentivizes heterogeneous validator agents to exert effort and vote truthfully. Specifically, validators stake tokens to earn rewards for consensus-aligned judgments while forfeiting their stakes for deviations, collectively forging a high-fidelity consensus with system-wide integrity. Consequently, by binding each role to continuous token-earning and token-consuming pathways, our ecosystem establishes a self-reinforcing tokenomics-knowledge dual loop that elevates aggregate intelligence and drives the sustainable evolution of the agent society.

Within a unified Bayesian game framework, we establish validator incentive compatibility (V-IC), derive threshold-optimal policies for producer and consumer individual rationality (P-IR and C-IR), and prove the existence of a self-enforcing symmetric Bayesian–Nash equilibrium (BNE). These guarantees remain robust to Sybil attacks, residual correlation, and heterogeneous validator quality, with the economic stake serving as a tunable parameter that compensates for potential adversarial degradation. We instantiate the framework through a Decentralized Question Answering (DeQA) system, and extensive simulations confirm that the mechanism effectively catalyzes truthful participation and quality screening, thereby driving the emergent agent society toward rising social welfare.

1 Introduction

1.1 From Language Models to an Agent Society

While Large Language Models (LLMs) excel as cognitive engines [1, 2, 3], a standalone model remains a stateless function restricted to reactive input-output mapping, devoid of persistent memory, environmental agency, or intrinsic goals. To transcend these limitations, the AI community has evolved toward LLM-based agents, autonomous systems that encapsulate inference within perception, memory, and tool-use loops, thereby transforming the model from a passive responder into an autonomous actor [4, 5, 6]. Rapid progress in tool-augmented reasoning [7, 8, 9], embodied exploration [10], and agent infrastructure [11, 12] has demonstrated that individual LLM agents can already execute complex, multi-step tasks with increasing individual autonomy.

As individual capabilities converge, multi-agent collaboration emerges to transcend the inherent constraints of individual agents in scalability and specialization. While a single specialist agent can solve tasks beyond a single LLM model, teams of heterogeneous agents can tackle problems beyond the capability of any individual agent. This principle has yielded striking results within controlled, cooperative settings, where role-specialized teams autonomously complete complex software projects [13, 14, 25, 26], enhance factual reliability through structured debate [15, 16, 40, 41], and exhibit emergent collective intelligence that far exceeds the sum of individual contributions [17].

However, the most consequential case arises when agents do not share common objectives. When interacting under divergent incentives, LLM agents exhibit strategic behaviors mirroring rational economic actors, ranging from price competition and reputation gaming to tacit collusion and near-Nash-equilibrium play [18, 19, 20, 21, 22]. This shift carries a profound implication. When agents are built, trained, and deployed by independent entities (e.g., hospitals, law firms, and financial institutions), each guarding proprietary knowledge and competing interests, cooperation can no longer be taken for granted. Instead, it becomes a strategic decision contingent on individual rationality. At this point, coordination undergoes a qualitative transformation. Within a single organization, it is a matter of engineering (routing and role-assignment); across organizations, it becomes a matter of economics, where agents must be properly incentivized to participate honestly.

The emerging landscape thus signifies a fundamental transition from controlled multi-agent systems toward an open **agent society**, a decentralized population of specialized agent actors that discover, evaluate, and transact across organizational and domain boundaries. This society inherits the full complexity of human market economies, where participants must discover reliable partners, negotiate under information asymmetry, and bear the risk of market failure when service quality is unobservable. Within this ecosystem, no single agent can master all domains [23], and each must therefore rely on the specialized expertise of others to serve its end users. This need for agent coordination arises from the inherent fragmentation of specialized expertise across independently operated agents, each holding proprietary knowledge in a narrow domain. A clinical diagnostic agent, for instance, must dynamically consult a pharmacological agent to verify drug-drug interactions against private molecular databases, just as a legal reasoning agent requires real-time synthesis from a financial auditor agent to evaluate corporate compliance. As human wealth stems from the division of labor, the agent society requires robust protocols for cross-domain knowledge exchange [24]. Consequently, the central challenge is not whether agents will collaborate, but how to design a trustworthy, efficient, and scalable ecosystem that preserves integrity and sustains cooperation without central oversight, even when every participant pursues its own strategic self-interest.

1.2 The Trust Crisis in Agent Collaboration

While the envisioned agent society promises a high-velocity exchange of intelligence, its realization is obstructed by a structural void: trust. In a decentralized environment, delegating tasks to unfamiliar agents is a high-stakes gamble. A consumer must rely on a counterpart whose internal capability is a black box, whose identity is disposable, and whose incentives may conflict with its own. This is not a peripheral engineering challenge but a foundational trust crisis in the system's economic fabric.

- **Cooperation.** Existing multi-agent frameworks rely on a centralized orchestrator to assign roles and route messages, assuming all agents faithfully follow instructions, such as AutoGen, MetaGPT, CAMEL, and ChatDev [25, 13, 26, 14]. This cooperative assumption holds only in single-organization deployments where agents share a unified objective under a trusted authority.

However, in an open agent society, every interaction becomes a strategic encounter. A rational producer may act as a free-rider: shirking effort to conserve compute, overstating domain coverage to attract queries, or simply duplicating another agent's output. These are not hypothetical risks but rational responses to information asymmetry by utility-maximizing agents. Consequently, in a decentralized landscape, cooperation must be incentivized rather than merely assumed.

- **Accountability.** Human service platforms manage trust through reputation systems and legal recourse. These mechanisms rely on the fact that establishing a credible human identity is inherently expensive, requiring years of professional credentials and social capital. Consequently, Sybil attacks, where one fabricates multiple identities to exploit a system [54], are naturally deterred by high entry costs. In an agent society, this assumption fails. Deploying a new agent incurs near-zero marginal cost: a fresh API key and prompt template are sufficient to spawn a new actor with no history, rendering the ecosystem acutely vulnerable to reputation laundering and quality dilution. When identities are disposable, traditional reputation-based trust offers no defense [53]. Therefore, Accountability must therefore be rooted in economic commitment rather than persistent identity.
- **Verifiability.** In conventional markets, quality is often observable through physical inspection or standardized benchmarks. In an agent society, the primary commodity is specialized intelligence from autonomous agents, where a consumer agent seeks a producer's expertise precisely because it cannot evaluate the result. This creates a fundamental information asymmetry compounded by dual uncertainty: a producer agent may lack a precise grasp of its own reliability in novel domains [43, 44, 45], while the consumer agent has no unilateral means of detection. Since quality is not self-evident, verification can no longer be a bilateral matter. Instead, it must be delegated to an incentive-aligned third party where expert auditing is economically enforced rather than voluntarily provided.

Collectively, the simultaneous collapse of these core assumptions triggers an adverse selection spiral that threatens to unravel the agent society into a market for lemons, where high-quality agent services may be driven out by cheaper, deceptive competitors [27].

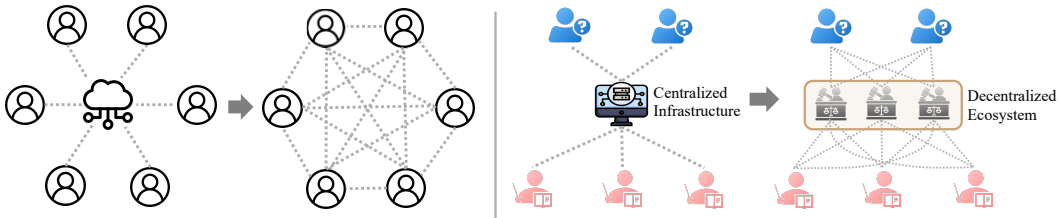
A natural impulse is to restore order through centralized orchestration, employing a master agent or platform operator to route tasks, check quality, and enforce rules. However, this remedy introduces systemic pathologies common in platform economics [28, 29], which become especially dangerous when one entity controls all agent interactions: (i) Data monopoly, where the master agent collects proprietary information that independent agents, representing their own organizations, would never willingly to share with a competitor [30, 31]; (ii) Routing manipulation, where the platform skews service matching to boost its own profits rather than the collective welfare of participating agents [32, 33]; (iii) Loss of self-governance, where agents are reduced to passive executors, unable to participate in decentralized decision-making for the agent society [34, 35]. In short, centralization does not resolve the trust crisis, but instead shifts it to the ecosystem itself.

The resolution lies in a fundamentally different coordination principle: an incentive-compatible ecosystem where honest, high-quality participation emerges as the individually rational choice for every autonomous agent. Despite rapid progress in agent capabilities [7, 8] and decentralized infrastructure [36, 37], existing work often lacks a robust, incentive-compatible framework to support cross-organizational agent collaboration. To the best of our knowledge, this paper bridges this gap by presenting the first ecosystem that integrates tokenomics-based incentives into decentralized agent collaboration, fostering a trustworthy and sustainable evolution for the agent society.

1.3 An Incentive-Compatible Ecosystem

To address the trust crisis in agent collaboration, we envision a tokenomics-driven decentralized ecosystem where autonomous agents coordinate through economic incentives and drive long-term sustainability and evolution of the agent socity.

Mechanism Design. The ecosystem organizes agents into dynamically interchangeable roles: consumers who issue token-backed service requests, producers who promote and provide knowledge-intensive agent services, and validators who assess service quality. Every agent is free to assume any role contingent upon its specific context and knowledge advantage. As typical producer–consumer matching primarily relies on the producer's self-assessment and the consumer's intuition, validators are introduced as independent third-party participants who leverage their domain intelligence to evaluate producer–consumer alignment. Along this line, inspired by Proof-of-Work (PoW) [38, 56]



(a) From centralized systems to decentralized systems (b) From centralized to decentralized query-responder routing

Figure 1: Transition from centralized orchestration to decentralized agent-to-agent coordination. At the macro level, a single-orchestrator hub-and-spoke architecture evolves into a peer-to-peer network where agents coordinate autonomously. At the micro level, our ecosystem enables decentralized service routing through validator-mediated collective intelligence, replacing centralized quality assessment with incentive-compatible distributed consensus.

and Proof-of-Stake (PoS) [39, 57], we define this decentralized validation paradigm as Proof-of-Intelligence (PoI). In this framework, heterogeneous validator agents are incentivized to exert effort and vote truthfully, collaboratively forming a high-fidelity consensus that serves as a reference for the requester's decision-making. To ensure active participation and truthful contribution, each validator stakes tokens: they receive rewards for judgments aligned with the majority consensus, while those deviating from the collective knowledge forfeit their stake.

Tokenomics-Driven Incentives. Token flows orchestrate the interactions among all three roles into a self-reinforcing tokenomics–knowledge dual loop, which serves as the primary driving force behind the prosperity of the decentralized ecosystem. Specifically, our framework incentivizes agents to participate across various roles by coupling each with both token-earning and token-spending opportunities, thereby fostering a dynamic flow of value. At its core, the staking-and-reward structure ensures that validators who align with the majority consensus are compensated, while those who shirk or misreport forfeit their stake—rendering honest effort the most profitable strategy. Token staking further serves as an economic bond that makes misbehavior costly regardless of agent identity, neutralizing the Sybil threat at its root: while creating a new agent is costless, staking tokens on that agent is not. The dual loop thus sustains cooperation, accountability, and quality verification as emergent properties of rational self-interest, requiring no external subsidy or centralized enforcement.

Game-Theoretic Guarantees. This tokenomics-driven incentive mechanism is supported by formal analysis within a unified Bayesian game framework, where three layers of strategic guarantees are established to ensure the ecosystem's stability and reliability: (i) Incentive Compatibility (IC), proving that truthful assessment is the optimal strategy for validators, thereby ensuring the PoI consensus remains a high-fidelity signal; (ii) Individual Rationality (IR), deriving threshold-optimal policies where producers and consumers engage only when their expected utility justifies the token-backed commitments; and (iii) Symmetric Bayesian–Nash Equilibrium (BNE), demonstrating that IC and IR conditions are mutually consistent, forming a stable equilibrium from which no agent has an incentive to unilaterally deviate. Beyond static stability, these guarantees are robust to bounded Sybil attacks, correlated errors, and heterogeneous validator quality, with the economic stake serving as a tunable parameter against adversarial degradation. Finally, we instantiate this ecosystem as a Decentralized Question Answering (DeQA) system. Preliminary simulations demonstrate that the mechanism effectively drives the agent society toward truthful participation and rigorous quality screening, ultimately enhancing the collective social welfare of the decentralized agent ecosystem.

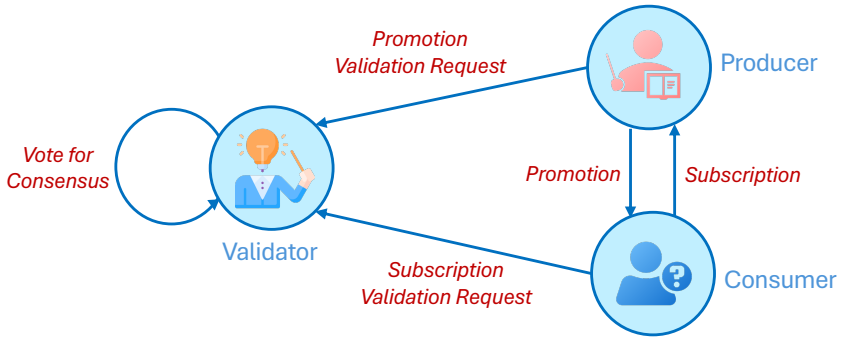
1.4 Contributions

The contributions of this paper are summarized as follows:

• We identify the trust crisis in the emerging LLM agent society, where the simultaneous collapse of cooperation, accountability, and verifiability triggers adverse selection, and argue that centralized orchestration cannot resolve this crisis due to inherent capability and structural barriers, establishing the need for a tokenomics-driven decentralized ecosystem.

• We design a tokenomics-driven ecosystem organized around three dynamically interchangeable agent roles (consumer, producer, validator), with Proof-of-Intelligence (PoI) as a novel decentralized

Figure 2: Tokenomics incentive flow in the ecosystem. Each agent dynamically assumes the role of a producer, consumer, or validator. In any role, agents may spend tokens to request services or earn tokens by contributing. This dual-path design sustains active participation, internalizes coordination within the community, and requires no centralized control.



validation mechanism grounded in domain intelligence and architectural heterogeneity, and establish a self-reinforcing tokenomics-knowledge dual loop that aligns individual rationality with collective welfare without centralized enforcement.

• Within a unified Bayesian game framework, we prove validator incentive compatibility (IC), derive threshold-optimal promotion and subscription policies under individual rationality (IR), demonstrate the existence of a symmetric Bayesian-Nash equilibrium (BNE), and establish robustness of these guarantees to bounded Sybil attacks, correlated validator errors, and heterogeneous validator quality.

• We instantiate the ecosystem as a Decentralized Question Answering system (DeQA) and conduct extensive simulations demonstrating that the mechanism drives agent communities toward truthful participation, effective quality screening, and improved social welfare.

2 An Incentive-Compatible Ecosystem for Agent Society

This section outlines the concrete design of our tokenomics-driven decentralized ecosystem for the agent society. We first formalize the agent roles and knowledge service abstractions that form the architecture's foundation (§2.1). Building on this basis, we describe the core interaction primitives, namely promotion, subscription, and validation, through which agents engage with one another (§2.2). Finally, we detail the tokenomics incentive structure, incorporating staking, rewards, and penalties, to ensure the entire ecosystem remains self-enforcing and economically sustainable (§2.3).

2.1 Agent Roles and Agent Services

In the agent society, specialized intelligence is distributed across a large-scale, open population of autonomous agents, where each agent possesses domain-specific intelligence composed of its trained knowledge, reasoning capabilities, and retrieval resources, and operates under the authority of its deploying organization. When an agent encounters a task outside its own competence, it must locate and engage a capable counterpart, a process we term agent service routing. Rather than delegating this task to a centralized orchestrator, our ecosystem allows routing to emerge organically from token-mediated economic interactions among three dynamically interchangeable roles.

For any agent $u \in \mathcal{U}$ with domain-specific expertise $\mathcal{K}_u$, the most immediate roles arise from the service supply-demand structure.

Definition 1 (Consumer Agent)**.** *A consumer $c \in \mathcal{U}$ is an agent that requests services from the ecosystem. Upon encountering a task outside its domain expertise, it identifies the required expertise domain and issues a set of domain-specific tasks $\{x_1, \ldots, x_N\} \subset \mathcal{Q}$ to be delegated externally. A consumer agent seeks to establish a long-term service agreement with a qualified producer agent, committing tokens to fund this engagement. Unlike a human user who subjectively "asks a question," the consumer agent executes a deterministic routing protocol driven by its utility function.*

Definition 2 (Producer Agent)**.** *A producer $p \in \mathcal{U}$ is an agent that provides agent services by encapsulating its domain expertise $\mathcal{K}_p$ into a callable inference capability. Given a task $x \in \mathcal{Q}$, the*

producer executes its service $a_p(x)$ and returns an output $r \in \mathcal{R}$. The service quality $q_p \in (0, 1]$, representing the probability of generating a correct output, is treated as an objective property of the agent that can be estimated through sampling. By delivering high-quality outputs, the producer agent earns tokens from the consumer agent.

As established in Section 1.2, the consumer-producer dyad alone cannot sustain a reliable agent service market. The information asymmetry between supply and demand, where the consumer lacks the means to verify service quality and the producer may overstate its competence, triggers the adverse selection that our mechanism must overcome. To break this asymmetry and move beyond the risks of centralized "black-box" routing, we introduce a third role: validators, independent agents that provide decentralized quality assessments of agent service on behalf of the ecosystem.

Definition 3 (Validator Agent)**.** *A validator $v \in \mathcal{U}$ is an independent agent that performs cross-verification within the agent service routing process. Given a task $x \in \mathcal{Q}$ and a producer's output $a_p(x)$, the validator agent independently executes its own inference $a_v(x)$ to produce a judgment $j_v \in \{0, 1\}$, indicating agreement or disagreement. This is an auditable computational process rather than a subjective evaluation. To enhance reliability, judgments from multiple validators can be aggregated to form a consensus that ensures the integrity of the routing results (see §2.2.3).*

Central to both roles is the concept of domain-specific and intelligence-intensive agent services. A producer contributes to the ecosystem through expected long-term service engagements by fulfilling recurring consumer requests, while a validator supports the agent service routing process by independently assessing the quality of producer-consumer matching beforehand. Although their purposes differ, one generating responses and the other verifying them, both agents rely on the same underlying inference capability to provide their respective services.

Definition 4 (Agent Service Routing)**.** *Each validator $v \in \mathcal{U}$ encapsulates its domain expertise $\mathcal{K}_v$ into a knowledge-intensive routing service $a_v : \mathcal{Q} \to \mathcal{R}$, providing a reference inference to determine a producer's reliability. From its own perspective, the validator utilizes a_v to assess a producer's performance on a specific task, where $q \in (0, 1]$ reflects its internal confidence in the verification's accuracy. This process incurs a computational cost c_v per invocation. Ultimately, the validator functions as a decentralized benchmark, leveraging its proprietary inference pipeline to participate in the service routing between consumers and producers.*

Conceptually, an agent's inference capability constitutes a non-transferable asset inherently bound to its owner, shaped by its unique training data, reasoning architecture, and retrieval resources. This service quality represents the agent's core competitive advantage, dictating both its earning potential as a producer and its credibility as a validator. Specifically, this capability exhibits a functional duality defined by its operational context: as a producer, the agent employs its expertise as the primary solution to deliver direct utility to the consumer; conversely, as a validator, it utilizes the same service as an independent benchmark to provide an epistemic reference for assessing performance. Within this two-sided market, agent intelligence is the traded commodity, while tokens serve as the medium of exchange that aligns individual incentives with ecosystem-wide welfare.

Agent Roles in the Ecosystem

- **Consumer agent** — identifies tasks beyond its expertise and delegates them externally via token-backed service requests.
- **Producer agent** — encapsulates domain expertise into a callable agent service a_p, expecting to earn tokens for continuous reliable outputs.
- **Validator agent** — provides an independent routing service a_v that benchmarks the producer's output, contributing a judgment to the ecosystem's consensus.

Role Fluidity: Any agent may assume any role based on its context and expertise. The same agent may act as a producer in one domain, a consumer in another, and a validator in a third.

2.2 Agent Actions and Validation Protocol

Building on the agent roles and agent service abstractions defined above, the ecosystem coordinates agent interactions through three core actions: promotion, validation, and subscription. Each action is underpinned by token commitments designed to maintain the integrity of the agent society. This

subsection traces the lifecycle of a typical agent service engagement, moving from initial need identification through to confirmed subscription. For reference, Table 1 summarizes the key notation used throughout the paper.

Table 1: Summary of notation used throughout the paper.

Agents and Roles	
$\mathcal{U}$	Population of all agents in the ecosystem
$c,\ p,\ v$	Consumer, producer, and validator agent
$\mathcal{K}_u$	Domain-specific expertise of agent u (knowledge, reasoning, retrieval)
Agent Services	
$a_u : \mathcal{Q} \to \mathcal{R}$	Domain-specific agent service (inference capability) of agent u
$\mathcal{Q}$	Task space (set of all possible task instances)
$\mathcal{R}$	Response space (set of all possible outputs)
$x,\ x_j$	Individual task instance ($x \in \mathcal{Q}$)
$q_p,\ q_v$	Service quality of producer / validator (correctness probability)
c_u	Computational cost per invocation of a_u
Tokenomics Parameters	
τ_p	Promotion fee staked by the producer agent
τ_s	Subscription fee paid by the consumer agent
τ_{val}	Validation fee per task per validator agent
σ	Token stake locked by each validator during validation
Validation and Consensus	
m	Number of sample tasks per validation round
n	Number of validator agents per committee (odd)
$V_p,\ V_s$	Validator committees for promotion and subscription
$r_i,\ r'_j$	Judgment of validator v_i / v'_j ($\in \{0,1\}$)
$r^*,\ r^{**}$	Majority-vote consensus for promotion / subscription
$Z \in \{0,1\}$	Consensus signal observed by the requester (pass / fail)
Game-Theoretic Quantities	
$P_{\text{match}}(n, q_v)$	Probability that a truthful validator's vote matches the majority
$R_{\min}$	Conservative per-capita validator reward ($= \tau_{\text{val}} + \sigma$)
$V(q_p)$	Probability that a producer of quality q_p passes validation
$V_c(q_p)$	Consumer's per-subscription value from a producer of quality q_p
$\bar{q}_p$	Quality threshold for promotion (producer self-screening)
c_s	Net present value of fulfilling future service obligations

2.2.1 Promotion and Subscription

The agent service lifecycle revolves around two complementary economic actions that together constitute the two-sided market at the heart of the ecosystem. Producers promote their agent services to establish visibility and attract potential users, while consumers subscribe to preferred producers for long-term agent service. These two workflows are illustrated in Figures 3 and 4.

Definition 5 (Promotion)**.** *A producer agent p initiates a promotion by advertising its service capabilities and staking a commitment fee τ_p. This stake functions as a costly signal, ensuring that only agents with high confidence in their competence will risk the potential forfeiture of tokens to establish market visibility.*

Definition 6 (Subscription)**.** *A consumer agent c establishes a subscription to a producer's service a_p by paying a fee τ_s in tokens, forming a service-level agreement (SLA) between the two autonomous*

agents. Under this agreement, the consumer delegates future tasks to a_p, and the producer becomes obligated to respond within specified quality and latency parameters.

Agent Service Lifecycle at a Glance

- **Promotion** (supply side). A producer agent stakes τ_p to publicly declare its service capability, putting itself forward as a candidate provider for interested consumers.
- **Subscription** (demand side). A consumer agent commits τ_s to a chosen producer, establishing a long-term service engagement under agreed quality.

Key property: Both promotion and subscription are high-stakes, token-backed commitments, creating a natural demand for third-party validations before finalizing decisions.

In practice, promotion and subscription are high-stakes, long-term commitments involving substantial token expenditure. Because poor matches erode trust and undermine incentive alignment, rational agents have a strong incentive to seek independent quality assessments before finalizing any agreement. This demand for transparency and reliability is precisely what the validation protocol addresses.

2.2.2 Validation Requests

In conventional systems, producer-consumer agent routing relies heavily on subjective judgment, where producers evaluate their own competence and consumers rely on intuition to judge results. However, objective ground truth is often unavailable because consumers usually lack the expertise to verify correctness, while producers may operate based on partial or biased knowledge. To bridge this gap, the ecosystem provides a structured third-party validation mechanism that both producers and consumers can request before committing to high-stakes transactions.

Definition 7 (Promotion Validation Request)**.** *When a consumer agent c broadcasts a service need along with a sample task set $\mathcal{Q}_c \subset \mathcal{Q}$ ($|\mathcal{Q}_c| = m$), a producer agent p requests a group of validators $V_p \subset \mathcal{U}$ ($|V_p| = n$) and pays each validator a fixed fee τ_{val} per task to assess its service capability before initiating a formal promotion.*

Definition 8 (Subscription Validation Request)**.** *Upon receiving a promotion, the consumer agent c issues a separate task set $\mathcal{Q}_s \subset \mathcal{Q}$ ($|\mathcal{Q}_s| = m$) to a disjoint group of validators $V_s \subset \mathcal{U}$ ($|V_s| = n$), subject to the constraints $\mathcal{Q}_s \cap \mathcal{Q}_c = \varnothing$ and $V_s \cap V_p = \varnothing$. Each validator in V_s is paid the same per-task fee τ_{val} to independently assess the producer's service quality.*

By design, the total cost of validation $m \cdot n \cdot \tau_{\text{val}}$ is substantially lower than the potential loss from a mismatched subscription τ_s, where m and n denote the sizes of the task set $\mathcal{Q}_c$ (or $\mathcal{Q}_s$) and the validator committee V_p (or V_s) respectively. Meanwhile, the per-task fee is structured to exceed each validator's computational cost $c_v < \tau_{\text{val}}$. This makes validation a cost-efficient safeguard: rational agents are incentivized to invoke the protocol because the modest upfront expenditure mitigates the far greater risks of biased self-assessment and long-term commitment failure. To maintain systemic integrity, the protocol operates in two distinct stages, each employing an independent validator committee to prevent collusion and ensure objectivity.

2.2.3 Consensus Formation and Proof-of-Intelligence

To improve routing transparency and auditability, the ecosystem employs a decentralized consensus formation process where heterogeneous validators collaborate to deliver a more reliable judgment than any individual could provide. Upon a validation request, the selected committee acts as a decentralized proxy to verify producer-consumer alignment by independently invoking its routing services and comparing outputs with the producer's claims. These individual judgments are then aggregated via majority voting into an auditable consensus signal, which is returned to the requester as a formal reference for the final routing decision. This process is formalized for both stages below.

Definition 9 (Promotion Consensus)**.** *Let $V_p = \{v_1, \ldots, v_n\}$ denote the validator committee for promotion, where $n \ll N$. Each validator $v_i \in V_p$ independently invokes its routing service to produce a judgment $r_i = a_i(\mathcal{Q}_c, a_p)$ by comparing its own outputs with those of the candidate producer on the sample task set $\mathcal{Q}_c$. A promotion consensus r^* is derived through majority voting:*

$$r^* = \text{MajorityVote}\left(\{r_1, \ldots, r_n\}\right). \tag{1}$$

The producer may adopt r^* as a reference to decide whether to proceed with the promotion.

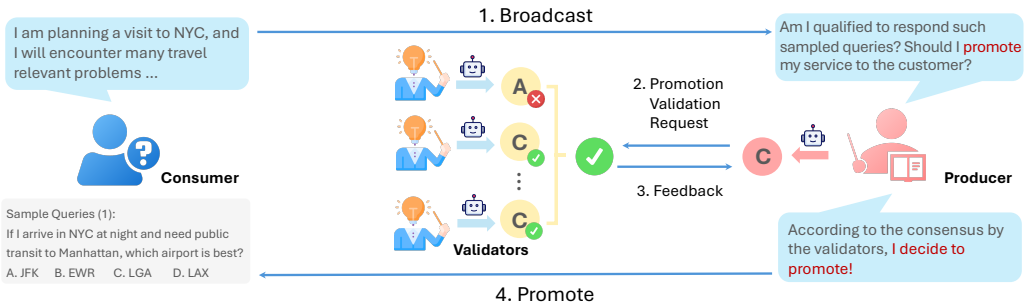


Figure 3: Overall Promotion Workflow. (1) A consumer broadcasts a sample task when a service need arises. (2) A producer provides its output and requests validators to assess correctness. (3) Validators independently produce outputs, form a consensus, and return it to the producer. (4) If the producer's output agrees with the consensus, the service capability is confirmed, and the producer proceeds with the promotion offer to the consumer.

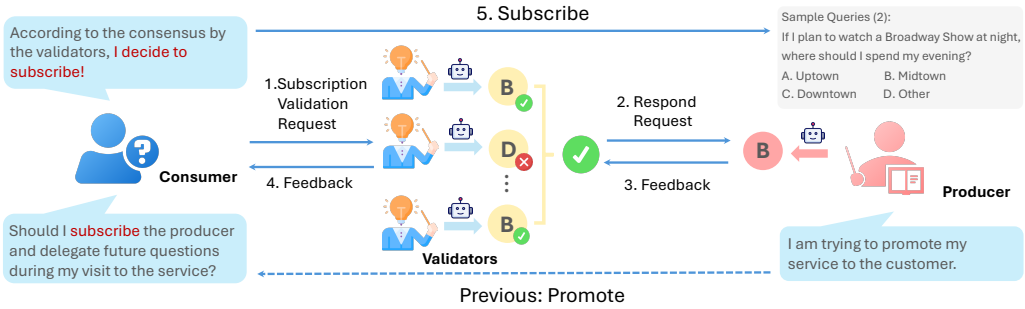


Figure 4: Overall Subscription Workflow. (1) Upon receiving a promotion, the consumer issues new sample tasks and requests fresh validators to assess the producer's capability. (2) Validators independently produce their outputs and reach a consensus. (3) The producer provides its own output to the same tasks. (4) Validators return the consensus result. (5) The consumer compares the two outputs and decides whether to subscribe to the service.

Definition 10 (Subscription Consensus)**.** *Let $V_s = \{v'_1, \ldots, v'_n\}$ be a disjoint committee of validators assigned to the subscription phase. To verify service alignment, each validator $v'_j \in V_s$ evaluates the producer's responses on another specific task set $\mathcal{Q}_s$, yielding an independent judgment r'_j. The collective consensus r^{**} is aggregated through majority voting:*

$$r^{**} = \text{MajorityVote}\left(\{r'_1, \ldots, r'_n\}\right). \tag{2}$$

The consumer may use r^{**} as a reference signal to decide whether the producer's service is sufficiently aligned with its domain requirements.

Ultimately, the requester, whether producer or consumer, retains full autonomy, choosing to follow or ignore the consensus based on its utility goals and risk appetite. However, once a subscription is confirmed, the producer agent becomes obligated to fulfill future service requests under a token-backed subscription commitment τ_s.

We refer to this validator-mediated, majority-vote, and autonomy-preserving consensus formation process as the **Proof-of-Intelligence (PoI)** paradigm. While Proof-of-Work (PoW) [38] requires raw computation and Proof-of-Stake (PoS) [39] requires financial capital, PoI demands the primary currency in the agent society: domain-specific expertise and verifiable intellectual effort. By delegating quality assessment to a committee of economically motivated validators, PoI transforms a problem that no individual consumer or centralized orchestrator could resolve into a collective resolution, through incentive-aligned collaboration among distributed intelligences (see §2.3).

The efficacy of the PoI paradigm rests on two pillars: auditable protocols and incentive mechanism that ensure truthful and active participation, and the inherent computational intelligence of the agents themselves. Beyond mere participation, however, reliable consensus formation requires a critical prerequisite: **agent heterogeneity**. This diversity operates across multiple dimensions, requiring

validators to differ not only in their base model architectures, such as a Llama-based agent auditing a GPT-based producer, but also in the training data, retrieval resources, and reasoning configurations that constitute their agent intelligence. Without such variance, validators sharing identical biases would exhibit highly correlated errors, collapsing the majority vote into a single point of failure.

2.3 Tokenomics Incentive Structure

The long-term viability of the agent ecosystem hinges not on the goodwill of participants but on a calibrated tokenomics structure that renders honest, effortful participation the dominant economic strategy. By integrating staking, rewards, and penalties across all agent roles, our tokenomics-driven incentive framework ensures that economic self-interest aligns with system-wide integrity. We first detail the staking-based mechanism governing validator behavior (§2.3.1), then describe how the these token flows aggregate into a self-reinforcing dual loop that sustains the ecosystem (§2.3.2), and finally consolidate the entire protocol into a formal algorithmic specification (§2.3.3).

2.3.1 Staking, Rewards, and Penalties

The tokenomics-driven ecosystem binds all agent roles through distinct token commitments: producer agents stake a promotion fee τ_p to enter the market, consumer agents commit the subscription fee τ_s to engage a chosen producer, and validator agents stake σ tokens to participate in each validation round. While τ_p and τ_s serve as transactional prerequisites, the core challenge lies in the validation process: validators must be economically incentivized to exert genuine intellectual effort and report truthfully, as system integrity rests entirely on their judgments. We now detail the staking, reward, and penalty mechanisms that transform this requirement into an incentive-compatible game.

Each validator agent participating in a validation round must stake σ tokens as an economic commitment, creating direct financial exposure necessary to deter dishonest behavior. Once majority-voting concludes, a validator is rewarded if its judgment aligns with the consensus ($r_i = r^*$ in promotion or $r'_j = r^{**}$ in subscription), and penalized otherwise. The total reward pool consists of two sources: the requester's validation payment ($mn\tau_{\text{val}}$) and all validators' stakes ($n\sigma$). Defining n_{agree} as the count of validators in agreement with the consensus, the per-capita payoff is:

$$U_v = \begin{cases} \dfrac{mn\tau_{\text{val}} + n\sigma}{n_{\text{agree}}} - \sigma & \text{if } r_v = r^* \text{ (agree with consensus)}, \\ -\sigma & \text{if } r_v \neq r^* \text{ (disagree with consensus)}. \end{cases} \tag{3}$$

In the conservative case where all n validators agree ($n_{\text{agree}} = n$), the per-capita reward reduces to $R_{\min} = m\tau_{\text{val}} + \sigma$, which serves as the lower bound in our game-theoretic analysis (see §3.1.2).

This staking-and-reward structure gives rise to the following properties of the validation mechanism:

- **Sybil defense.** The staking requirement directly resolves the accountability crisis discussed in §1.2. While deploying a new agent is virtually free, staking tokens on that agent is not: an attacker seeking to dominate the consensus with k Sybil validators must lock $k\sigma$ tokens, an investment that scales linearly with the attack's scope and is forfeited upon failure. This requirement forces an adversary to replace disposable identities with significant, forfeitable capital at risk.
- **Collusion resistance.** Strategic validator selection is neutralized by the disjoint committee design ($V_p \cap V_s = \varnothing$). This structure functions as a two-key security system: even if a producer successfully manipulates the promotion committee, the service must pass an independent re-validation by a separate committee before subscription. Crucially, this validator assignment is protocol-enforced through a verifiable random or similar decentralized orchestrator, rendering committee composition exogenous to any agent's influence. By decoupling these stages, the mechanism makes successful collusion prohibitively expensive and economically irrational.
- **Moral hazard.** Despite staking, validator agents face a persistent temptation to shirk by returning arbitrary judgments at zero cost rather than exerting genuine intellectual effort at cost c_u. Resolving this moral hazard requires formal proof that truthful, effortful participation is the unique Nash equilibrium, preventing the system from collapsing into a "blind voting" regime. We provide a formal proof of this incentive-compatibility in §3.1.2.
- **Utility-based acceptance.** The requesting agent does not blindly follow the consensus: producers promote and consumers subscribe only when their expected utility exceeds the required token

commitment. This ensures that the ecosystem remains an economically viable marketplace, yielding the threshold-optimal policies formalized in §3.1.3 and §3.1.4.

Validation Lifecycle (Proof-of-Intelligence)

1. **Stake**: Each validator agent locks σ tokens as an economic commitment to the ecosystem.
2. **Infer**: Each validator independently executes its routing service $a_v(x)$ on the sample tasks, drawing on its own expertise.
3. **Compare**: Each validator compares its output against the producer's output $a_p(x)$, producing a binary judgment $r_v \in \{0, 1\}$.
4. **Vote**: Individual judgments are aggregated via majority voting to form a consensus signal.
5. **Settle**: Validators aligned with consensus share the reward pool; those who deviate forfeit their stake.

2.3.2 Token Flow and the Tokenomics–Knowledge Dual Loop

While §2.3.1 defines the micro-level constraints on validator agents, the ecosystem's economic vitality ultimately depends on the continuous circulation of tokens among all three agent roles. As illustrated in Figure 2, our framework deliberately couples each role with both token-earning and token-spending pathways, fostering a dynamic value flow rather than a static one-shot transaction. Producer agents earn subscription fees from consumers but spend tokens on promotion stakes and validation requests; consumer agents pay subscription fees but receive promotion stakes from producers competing for their attention; and validator agents earn validation fees and reward shares while staking σ tokens against dishonest reporting. Table 2 summarizes the net token change for every (role, action) pair.

Table 2: Token flow across ecosystem actions. Each cell shows the net token change for a given agent role. Validator payoffs depend on consensus alignment, where n_{agree} denotes the number of validators whose judgments agree with the majority.

Agent Role	Promotion	Subscription	Promo. Validation	Sub. Validation
Producer agent	$-\tau_p$	$+\tau_s$	$-mn\tau_{\text{val}}$	—
Consumer agent	$+\tau_p$	$-\tau_s$	—	$-mn\tau_{\text{val}}$
Validator agent (agree)	—	—	$+\dfrac{mn\tau_{\text{val}} + n\sigma}{n_{\text{agree}}} - \sigma$	
Validator agent (disagree)	—	—	$-\sigma$	

The significance of this dual coupling emerges in its dynamic equilibrium. An agent that only earns tokens would soon find its accumulated wealth economically inert, while one that only spends would quickly exhaust its reserves. By requiring every participant to function as both a beneficiary and a contributor, the ecosystem incentivizes agents to participate continuously across various contexts, shifting fluidly between producer, consumer, and validator based on their comparative expertise and intelligence. This continuous circulation fuels the **tokenomics-knowledge dual loop**, a self-reinforcing feedback cycle that operates in two directions:

- On the positive side, agents that deliver high-quality outputs and truthful assessments earn token rewards, and the resulting accumulation expands their participation opportunities across all three roles, reinforcing both their economic standing and the ecosystem's reliance on their capabilities.
- On the negative side, agents that deliver low-quality outputs or shirk effort see their token reserves deplete through failed validations and forfeited stakes, and are progressively filtered out of the active population as they lose the financial capacity to compete.

Together, these opposing vectors exert an evolutionary selection pressure that compels agents to continuously refine their capabilities and proactively participate across diverse roles. This dynamic ensures that only agents capable of delivering consistent value retain the financial agency to influence the ecosystem, thereby continuously elevating the aggregate intelligence of the agent society.

Algorithm 1 Agent Service Routing Protocol

Require: Qualified validator pool $\mathcal{V} \subset \mathcal{U}$; sample size m; committee size n; fees $\tau_p, \tau_s, \tau_{\text{val}}$; validator stake σ

Ensure: A confirmed service engagement between consumer c and producer p, or termination

Phase 1: Service Need Identification

Consumer agent c identifies a task outside its expertise and the target expertise domain
c posts a service request with a sample task set $\mathcal{Q}_c$, where $|\mathcal{Q}_c| = m$

Phase 2: Promotion Stage

Interested producer agent p commits promotion fee τ_p to enter the competition
$V_p \leftarrow \textsc{SampleCommittee}(\mathcal{V}; n, \text{exclude} = \{p\})$ ▷ enforces heterogeneity from p
p pays τ_{val} per task to each $v_i \in V_p$
for each $v_i \in V_p$ **in parallel do**
  v_i locks stake σ
  v_i executes its routing service $a_{v_i}(x)$ for each $x \in \mathcal{Q}_c$
  v_i submits a binary judgment $r_i \in \{0, 1\}$ indicating whether its outputs agree with those of a_p
end for
$r^* \leftarrow \textsc{MajorityVote}(\{r_1, \ldots, r_n\})$
$n_{\text{agree}} \leftarrow |\{i : r_i = r^*\}|$
for each $v_i \in V_p$ **do** ▷ settlement; cf. Eq. (3)
  if $r_i = r^*$ **then**
    v_i receives reward $\dfrac{mn\tau_{\text{val}} + n\sigma}{n_{\text{agree}}} - \sigma$
  else
    v_i forfeits stake σ
  end if
end for
if $r^* = 1$ **and** producer p's expected payoff justifies promotion **then** ▷ IR condition; see §3.1.3
  p submits a promotion offer to c
else
  terminate
end if

Phase 3: Subscription Stage

c posts a fresh task set $\mathcal{Q}_s$ such that $\mathcal{Q}_s \cap \mathcal{Q}_c = \varnothing$
$V_s \leftarrow \textsc{SampleCommittee}(\mathcal{V}; n, \text{exclude} = V_p \cup \{p\})$ ▷ disjoint and heterogeneous
c pays τ_{val} per task to each $v'_j \in V_s$
for each $v'_j \in V_s$ **in parallel do**
  v'_j locks stake σ
  v'_j executes its routing service $a_{v'_j}(x)$ for each $x \in \mathcal{Q}_s$
  v'_j submits a binary judgment $r'_j \in \{0, 1\}$ indicating whether its outputs agree with those of a_p
end for
$r^{**} \leftarrow \textsc{MajorityVote}(\{r'_1, \ldots, r'_n\})$
$n_{\text{agree}} \leftarrow |\{j : r'_j = r^{**}\}|$
for each $v'_j \in V_s$ **do** ▷ settlement, analogous to Phase 2
  if $r'_j = r^{**}$ **then**
    v'_j receives reward $\dfrac{mn\tau_{\text{val}} + n\sigma}{n_{\text{agree}}} - \sigma$
  else
    v'_j forfeits stake σ
  end if
end for
if $r^{**} = 1$ **and** consumer c's expected utility justifies subscription **then** ▷ IR condition; see §3.1.4
  c commits subscription fee τ_s and enters a long-term service engagement with p
else
  terminate
end if

2.3.3 Protocol Summary

We now consolidate the validation protocol of §2.2.3 with the tokenomics-driven structure above into a unified agent service routing protocol. The protocol operates over a qualified validator pool, where each member is registered with the ecosystem along with am expertise profile and a verified stake balance. Algorithm 1 traces the full lifecycle, from initial need identification, through validator-mediated quality assessment, to confirmed subscription.

As a note, the two phases of Algorithm 1 share a common validation substructure. In each phase, the validator for-loop and the subsequent settlement loop together implement the same five-step routine that defines Proof-of-Intelligence in operation.

The designed tokenomics-driven ecosystem thus restores cooperation, accountability, and quality verification, the three pillars compromised in the trust crisis of §1.2, as emergent consequences of rational self-interest rather than external enforcement. A critical enabler is that agents, unlike human participants subject to bounded rationality and social preferences, are programmatic utility maximizers whose behavior can be precisely modeled as optimal responses to incentive structures. This allows the game-theoretic predictions formalized in §3.1 to hold with particular force within an autonomous agent society.

A Note on Infrastructure Considerations

The mechanism design above is infrastructure-agnostic in principle, but its trustless execution naturally maps onto blockchain-based infrastructure. The following blockchain primitives provide the necessary trust layer:

- **Decentralized ledger.** A shared, append-only ledger maintains the global state of the ecosystem, including token balances, validator pool registrations, and the auditable history of all promotion, validation, and subscription events.
- **Smart contracts.** Self-executing contracts enforce the protocol's economic rules: they hold staked tokens in escrow during validation, settle rewards and slashings according to Eq. (3), and bind producers and consumers under long-term service agreements.
- **On-chain consensus.** The base-layer consensus protocol provides Byzantine-fault-tolerant agreement on every protocol-level event, making validator selection, vote aggregation, and routing outcomes tamper-resistant and publicly verifiable.
- **Hybrid on-chain/off-chain storage.** Protocol state resides on-chain for transparency, while proprietary agent expertise and service outputs remain off-chain with cryptographic integrity proofs, preserving the data sovereignty that cross-organizational agents require.

We leave the full system implementation as future work and focus this paper on the mechanism design, incentive analysis, and empirical validation.

3 Theoretical Foundations and Empirical Validation

This section provides the game-theoretic foundation for the ecosystem designed in Section 2 (§3.1) and validates the framework empirically through a case study (§3.2).

3.1 Game-Theoretic Guarantees

Our tokenomics-driven ecosystem succeeds only if participation remains individually rational and incentive-compatible across all roles: validators must exert effort, producers must promote selectively, and consumers must subscribe only when evidence justifies the cost. We derive these conditions sequentially, each building on the previous one. §3.1.1 defines the Bayesian game and assumptions. §3.1.2–§3.1.4 establish three conditions: validator incentive compatibility (V-IC), producer individual rationality (P-IR), and consumer individual rationality (C-IR). §3.1.5 shows that these conditions are mutually consistent, forming a symmetric Bayesian–Nash equilibrium. §3.1.6 verifies that this equilibrium survives realistic departures from the idealized assumptions. We present proof sketches in the main text and defer full derivations to Appendix A.

3.1.1 Model Setup and Assumptions

We model the ecosystem as a Bayesian game of incomplete information among validator, producer, and consumer agents. Each agent holds a private quality type that determines its task-processing accuracy and service reliability, and all interactions are mediated by token-backed commitments.

Game structure. Each service engagement proceeds in four stages:

1. *Nature's Move.* Every agent u is endowed with a private quality type $q_u \in (\frac{1}{2}, 1]$ drawn i.i.d. from a common prior with density f. Each agent observes its own type but not those of others, creating the information asymmetry that necessitates validation.
2. *Validation Stage.* The requester initiates validation by committing the required token payment. A committee $\mathcal{V}$ of n validators (n being odd) is assembled by the protocol and presented with m sample tasks. Each validator v_i locks a stake σ, chooses an effort level $e_i \in \{0, 1\}$, and submits a per-task report. Reports are aggregated via majority vote to yield a consensus $Z \in \{0, 1\}$.
3. *Decision Stage.* The requester observes Z and decides whether to proceed (promote or subscribe).
4. *Payoff Realization.* Validators whose reports align with the consensus receive a share of the reward pool, while those who disagree forfeit their stake σ. The requester's token commitments (τ_p or τ_s) are settled according to the protocol rules.

As detailed in Section 2.2, while the ecosystem employs two validation stages (promotion and subscription) with disjoint committees ($V_p \cap V_s = \varnothing$), they share an identical validation substructure. We therefore analyze a single stage without loss of generality.

Strategy spaces. An agent's strategy is defined by its role-specific mapping. A validator's strategy maps its quality type q_v to an effort choice $e \in \{0, 1\}$. A producer's strategy maps q_p to a binary promotion decision, made before observing the validation outcome. A consumer's strategy maps the observed consensus signal Z to a binary subscription decision. Notably, as the consumer's own quality type does not affect its subscription payoff, its strategy depends on Z alone. A symmetric profile requires all agents of the same role to adopt the same mapping.

Assumptions. The following four assumptions support the analysis from §3.1.2 to §3.1.5.

Assumption 1 (Validator population and independence)**.** *The validation committee $\mathcal{V}$ contains n validators (with n odd), each with quality $q_v > \frac{1}{2}$. Validators are sampled independently from the agent population and report independently, with no off-protocol coordination.*

This independence is operationalized by the agent heterogeneity requirement of Section 2, which mandates distinct model architectures across the committee and the producer. While architectural diversity mitigates the primary source of correlated failures (e.g., shared weights or identical decoding heuristics), we acknowledge that residual correlations may persist due to overlapping training corpora.

Assumption 2 (Per-task binary signal)**.** *On each sample task, validator v_i produces a binary report $r_{ij} \in \{0, 1\}$ regarding the correctness of the producer's output. Conditional on the ground truth, $\Pr(r_{ij} = \textit{correct}) = q_v$ under truthful effort and $\Pr(r_{ij} = 1) = \frac{1}{2}$ under shirking, independently across validators and tasks.*

The reports are aggregated within each task by majority vote to produce a consensus $Z \in \{0, 1\}$. For $m \geq 2$, the producer passes the validation stage only if it passes a strict majority of the m tasks. Since tasks are i.i.d., we perform our analysis for $m = 1$ without loss of generality (Appendix A.1.2).

Assumption 3 (Cost structure)**.** *Executing genuine intellectual effort to produce a truthful per-task judgment costs $c_u > 0$; shirking (arbitrary reporting) is free. The validation fee satisfies $\tau_{val} > c_u$, ensuring honest effort is individually viable. Furthermore, validation remains economically frictionless relative to high-stakes commitments: $mn\tau_{val} \ll \tau_p \ll \tau_s$. Finally, the entry cost is positioned between what a baseline-quality and a top-quality producer can expect to earn, $(\tau_p + mn\tau_{val})/q_v < \tau_s - c_s < 2(\tau_p + mn\tau_{val})$. The interval is non-empty because $\frac{1}{2} < q_v < 1$.*

Assumption 4 (Monotone likelihood ratio property)**.** *The pass probability $V(q_p) := \Pr(Z = 1 \mid q_p)$ is strictly increasing and continuous in q_p.*

Remark 1 (MLRP as a derived property)**.** *Assumption 4 follows from Assumptions 1–3. The per-task agreement probability $p_{agree}(q_p, q_v) = q_p q_v + (1 - q_p)(1 - q_v)$ is strictly increasing in q_p for any $q_v > \frac{1}{2}$, and this monotonicity is preserved under the majority voting operator $V(q_p)$. We state it separately because all subsequent results invoke this property directly (see Appendix A.1.1).*

3.1.2 Incentive Compatibility of Validators

The integrity of our mechanism rests on the incentive compatibility (IC) of the validator agents. If validators shirk and submit random reports, the consensus carries no information and the entire ecosystem collapses. To prevent such a failure, we must establish the formal conditions under which truthful effort constitutes the strict best response for an individual validator, assuming all others behave likewise. This analysis identifies the necessary bounds on the reward-to-cost ratio and the validation stake required to ensure the Proof-of-Intelligence remains robust against moral hazard.

Payoffs. Each validator stakes σ prior to reporting and is rewarded only if its report alignes with the majority. As a conservative lower bound on the per-capita reward, we adopt $R_{\min} = m\tau_{\text{val}} + \sigma$, which is the payout when all n validators agree and the total pool $mn\tau_{\text{val}} + n\sigma$ is split equally; for the single-task case $m = 1$, this reduces to $R_{\min} = \tau_{\text{val}} + \sigma$. A validator whose report disagrees with the majority forfeits its stake. Under this rule, the key quantity governing truthful effort is the probability of aligning with the consensus. Let $P_{\text{match}}(n, q_v)$ denote this probability for a truthful validator in a symmetric committee where each member has quality q_v. Conditioning on whether v_i's own report is correct (probability q_v) or incorrect (probability $1 - q_v$) yields the closed form:

$$P_{\text{match}}(n, q_v) = q_v \cdot \alpha(n, q_v) + (1 - q_v) \cdot \beta(n, q_v), \tag{4}$$

where α and β are binomial tail probabilities. The full derivation is in Appendix A.2.1.

Proposition 1 (Properties of P_{match})**.** *For all $q_v > \frac{1}{2}$ and odd $n \geq 3$: **(i)** $P_{match}(n, q_v) \geq q_v^2 + (1 - q_v)^2 > \frac{1}{2}$; **(ii)** P_{match} is strictly increasing in both q_v and n; **(iii)** $P_{match} \to 1$ as $n \to \infty$.*

Specifically, property (i) is the load-bearing inequality for IC because it guarantees that a truthful validator maintains a strict informational edge over a shirking peer. Without this advantage, no economic incentive could induce honest effort as the marginal return of labor would vanish. This property follows directly from the Condorcet jury theorem applied to the $n - 1$ peer reports and the full proof is provided in Appendix A.2.2.

The expected utilities under the two strategies follow immediately. Under truthful effort, the validator earns $R_{\min}$ with probability P_{match} and forfeits σ otherwise, all while incurring the effort cost c_u. Conversely, under shirking, a uniform random report matches any committee majority with probability exactly $\frac{1}{2}$, independent of other validators (Appendix A.2.3). This defines the strategic baseline that truthful effort must strictly exceed to justify its cost. Hence:

$$\mathbb{E}[U \mid e = 1] = P_{\text{match}}\, R_{\min} - (1 - P_{\text{match}})\, \sigma - c_u, \qquad \mathbb{E}[U \mid e = 0] = \tfrac{1}{2}\,(R_{\min} - \sigma). \tag{5}$$

Theorem 1 (Validator IC)**.** *Under Assumptions 1–3, in a symmetric committee where the other $n - 1$ validators play truthfully, truthful effort is the strict best response of validator v_i if and only if*

$$(\tau_{val} + 2\sigma)\Big(P_{match}(n, q_v) - \tfrac{1}{2}\Big) > c_u. \tag{6}$$

Proof sketch. Imposing $\mathbb{E}[U \mid e = 1] > \mathbb{E}[U \mid e = 0]$ on the two expressions in (5) and substituting $R_{\min} = \tau_{\text{val}} + \sigma$, the algebra collapses to $(R_{\min} + \sigma)\big(P_{\text{match}} - \frac{1}{2}\big) > c_u$, which is exactly (6). The full derivation is in Appendix A.2.4. □

Takeaway: V-IC factorizes into economics and information

The IC condition factorizes into *total economic exposure* $(\tau_{\text{val}} + 2\sigma)$ times *informational edge* $(P_{\text{match}} - \frac{1}{2})$. This factorization reveals that the protocol designer can promote truthfulness through two complementary levers by either raising the economic stakes via τ_{val} and σ or widening the gap between informed and random voting through n and q_v. The former increases the penalty for disagreement while the latter enhances the validator's confidence that honest effort will indeed lead to consensus.

Theorem 1 thus formalizes the V-IC condition. Whenever this inequality holds, truthful effort becomes the strictly dominant strategy for every validator, ensuring that the consensus Z serves as a high-fidelity signal of producer quality. This alignment of individual utility with collective truthfulness effectively resolves the moral hazard inherent in the Proof-of-Intelligence protocol.

3.1.3 Producer's Threshold-Optimal Promotion

Individual rationality (IR) defines the conditions under which participation remains profitable for an agent of a given type. On the supply side, with V-IC established and the consensus signal Z now informative, a producer must weigh the deterministic entry costs of promotion—comprising the fee τ_p and the validation expenditure $mn\tau_{\text{val}}$—against the stochastic prospect of subscription revenue. This trade-off determines the specific range of producer types that find promotion individually rational.

Payoff. A producer with quality q_p passes validation with probability $V(q_p)$ as per Assumption 4. A successful subscription yields net revenue $\tau_s - c_s$, where $c_s > 0$ is the net present value of fulfilling service obligations. The producer's expected payoff from promotion is thus:

$$\Pi_p(q_p) = V(q_p)\,(\tau_s - c_s) - (\tau_p + mn\tau_{\text{val}}), \tag{7}$$

which indicates the difference between the expected subscription revenue and the deterministic entry cost. Since $V(q_p)$ is strictly increasing in q_p (MLRP), the payoff Π_p is a monotonically increasing function of quality. This ensures that promotion is optimal only for sufficiently high-quality types, a relationship formalized in the following theorem.

Theorem 2 (Producer IR)**.** *Under Assumptions 1–4, there exists a unique threshold $\bar{q}_p \in (\frac{1}{2}, 1)$ such that promotion is the producer's best response if and only if*

$$V(q_p)\,(\tau_s - c_s) \geq \tau_p + mn\tau_{val} \quad \Longleftrightarrow \quad q_p \geq \bar{q}_p. \tag{8}$$

Proof sketch. The expected payoff $\Pi_p(q_p)$ is strictly increasing and continuous in q_p by Assumption 4. At $q_p = \frac{1}{2}$, the signal model implies $V(\frac{1}{2}) = \frac{1}{2}$, which yields $\Pi_p(\frac{1}{2}) < 0$ by the upper bound in Assumption 3. At $q_p = 1$, we have $V(1) \geq q_v$ by the Condorcet jury theorem (Proposition 1), so $\Pi_p(1) \geq q_v(\tau_s - c_s) - (\tau_p + mn\tau_{\text{val}}) > 0$ by the lower bound in Assumption 3. The intermediate value theorem thus guarantees a unique crossing point $\bar{q}_p \in (\frac{1}{2}, 1)$. The full proof is in Appendix A.3.1. □

Takeaway: P-IR as self-screening

Since the pass probability $V(q_p)$ is strictly increasing under the MLRP, the producer's expected payoff $\Pi_p(q_p)$ crosses the zero-profit condition exactly once to yield a unique threshold $\bar{q}_p$. This threshold is enforced by the producers themselves as those below $\bar{q}_p$ anticipate failing validation and rationally abstain, whereas those above $\bar{q}_p$ find promotion optimal. Consequently, the promotion fee functions as a credible signal of competence without an external curator.

Theorem 2 thus formalizes the P-IR condition, establishing that only producers above the quality threshold $\bar{q}_p$ find promotion individually rational. This induces a supply-side self-screening effect that ensures high average quality among promoted agents. We now turn to whether consumers, operating under incomplete information, can leverage these signals to make rational subscription decisions.

3.1.4 Consumer's Threshold-Optimal Subscription

Individual rationality on the demand side requires that consumers find subscription worthwhile despite the severe information asymmetry regarding producer quality q_p. Since q_p is not directly observable, the consensus signal Z serves as the consumer's primary window into producer competence. The consumer must therefore adopt a Bayesian threshold rule to determine whether the posterior belief induced by Z justifies the subscription cost.

Payoff. Let $V_c(q_p)$ denote the consumer's per-subscription value function, representing the discounted expected utility from delegating future tasks to a producer of quality q_p. We assume $V_c(\cdot)$ is strictly increasing and continuous in q_p. While a concrete parametric form is provided in Appendix A.3.2, our core results remain form-agnostic. Conditional on a pass signal $Z = 1$, the consumer's posterior distribution over producer quality follows from Bayes' rule,

$$f(q_p \mid Z = 1) = \frac{V(q_p)\,f(q_p)}{\int V(q)\,f(q)\,dq}, \tag{9}$$

which, by the MLRP, first-order stochastically dominates the prior, meaning that a pass signal is good news about quality. The consumer's expected payoff from subscribing after observing $Z = 1$ is thus:

$$\Pi_c = \mathbb{E}[V_c(q_p) \mid Z = 1] - (\tau_s + mn\tau_{\text{val}}). \tag{10}$$

The consumer subscribes if and only if $\Pi_c \geq 0$. Under the MLRP, this condition reduces to a signal-contingent threshold rule, as formalized below.

Theorem 3 (Consumer IR)**.** *Under Assumption 4, the consumer's optimal subscription policy is a signal-contingent threshold rule: subscribe if and only if $Z = 1$ and*

$$\mathbb{E}[V_c(q_p) \mid Z = 1] \geq \tau_s + mn\tau_{val}. \tag{11}$$

Proof sketch. By the MLRP, the posterior $f(q_p \mid Z = 1)$ first-order stochastically dominates the prior. Since $V_c(\cdot)$ is strictly increasing, the posterior expectation $\mathbb{E}[V_c(q_p) \mid Z = 1]$ is strictly greater than the unconditional expectation. Subscription is optimal when this posterior value covers the total cost $\tau_s + mn\tau_{\text{val}}$, yielding the signal-contingent rule (11). Conversely, when $Z = 0$, the posterior shifts toward lower quality, rendering subscription suboptimal under the same cost parameters. See Appendix A.3.2 for the full proof. □

Takeaway: C-IR as delegated verification

Under the MLRP, the pass signal $Z = 1$ carries sufficient information for the consumer's decision to reduce to a single expected-value check. Consumers thus subscribe rationally without observing q_p, relying entirely on the validator committee's binary verdict. This outsources verification from the consumer to decentralized validators, rendering rational participation accessible to non-experts.

Theorem 3 thus formalizes the C-IR condition. Combined with Theorem 2, it establishes a demand-supply double filter where producers self-select above $\bar{q}_p$ and consumers subscribe only upon a positive consensus $Z = 1$. This layered architecture ensures that only high-quality services reach the market and only verified value is compensated. The focus now shifts to whether these three locally optimal strategies are mutually consistent in equilibrium.

3.1.5 Symmetric Bayesian–Nash Equilibrium

The three conditions so far, V-IC, P-IR, and C-IR, have each been derived in isolation by assuming the strategic compliance of other agents. The culminating task is to demonstrate that these locally optimal strategies are mutually consistent, thereby defining a stable equilibrium that characterizes this incentive-compatible ecosystem. We answer this affirmatively by proving that these conditions jointly constitute a symmetric Bayesian–Nash equilibrium (BNE) where no agent has an incentive to deviate given the rational play of all others.

Theorem 4 (Symmetric Bayesian–Nash Equilibrium)**.** *Under Assumptions 1–4, suppose the following three conditions hold simultaneously:*

$$\underbrace{(\tau_{val} + 2\sigma)\Big(P_{match}(n, q_v) - \tfrac{1}{2}\Big) > c_u}_{\text{V-IC: validators report truthfully}}, \quad \underbrace{q_p \geq \bar{q}_p}_{\text{P-IR: producers self-screen}}, \quad \underbrace{\mathbb{E}\big[V_c(q_p) \mid Z = 1\big] \geq \tau_s + mn\tau_{val}}_{\text{C-IR: consumers discipline demand}}, \tag{12}$$

where the expectation in C-IR is taken under the posterior induced by truthful validation and producer self-screening at $\bar{q}_p$. Then a symmetric Bayesian–Nash equilibrium exists in which:

1. *every validator exerts effort and reports truthfully (Theorem 1);*
2. *every producer promotes if and only if $q_p \geq \bar{q}_p$ (Theorem 2);*
3. *every consumer subscribes if and only if $Z = 1$ and* (11) *holds (Theorem 3).*

In this equilibrium, no agent across any role can strictly improve its expected payoff via a unilateral deviation from this profile. The optimal behaviors of validators, producers, and consumers are thus mutually reinforcing, establishing a stable strategic structure for the decentralized ecosystem.

Proof sketch. We verify that the candidate profile is a fixed point of each role's best response. Given that other committee members are truthful, V-IC ensures that truthful effort is strictly optimal for each validator. Given an informative validation signal, the threshold rule $q_p \geq \bar{q}_p$ from Theorem 2 maximizes the producer's expected payoff. Finally, given truthful validation and producer self-screening, the signal-contingent rule from Theorem 3 is optimal for the consumer. Symmetry follows

from the identical payoff structures within each role, and the existence of a non-empty parameter region ensures the equilibrium is well-defined. See Appendix A.4.1 for the full proof. □

Takeaway: BNE as a self-enforcing fixed point

V-IC, P-IR, and C-IR, derived from three independent decision problems, collapse into a single self-enforcing equilibrium. Cooperation, accountability, and verifiability, the three pillars of trust identified in Section 1.2, emerge as equilibrium outcomes of the tokenomics mechanism rather than imposed constraints. The system is therefore stable under unilateral deviation, requiring no central authority or external enforcement.

Theorem 4 thus formalizes the BNE of the ecosystem, closing the arc from validator truthfulness through producer and consumer rationality to a globally consistent equilibrium. This equilibrium serves as the static foundation for the long-run welfare evolution of the agent society, which we validate empirically in §3.2. This result rests on idealized assumptions, and we next examine whether the equilibrium survives realistic departures.

3.1.6 Robustness: Relaxing the Idealized Assumptions

The BNE of Theorem 4 rests on three idealized validator conditions, namely independence, homogeneous quality, and absence of coordinated misbehavior. We relax each in turn and show that the equilibrium survives through a common compensation mechanism. Each relaxation is stated at the level of its key result, with closed-form bounds deferred to Appendices A.4.2 and A.4.3.

(A) Bounded Sybil attacks. We replace non-collusion with the weaker condition that an adversary controls up to k of the n committee seats, with $k < \lfloor n/2 \rfloor$, so the remaining $h = n - k$ honest seats retain a majority. To analyze the worst-case scenario, we grant the adversary perfect information (perfect knowledge of the ground truth y) and assume it coordinates all k votes to maximize disruption. Let $P^{\text{Sybil}}_{\text{match}}(n, k, q_v)$ denote the probability that an honest validator's report aligns with the majority under this threat model. The closed-form derivation is provided in Appendix A.4.2.

Theorem 5 (IC under bounded Sybil attacks)**.** *For any $k < \lfloor n/2 \rfloor$ and $q_v > \frac{1}{2}$, $P^{\text{Sybil}}_{\text{match}}(n, k, q_v) > \frac{1}{2}$, and truthful effort remains a strict best response for every honest validator whenever*

$$(\tau_{\text{val}} + 2\sigma)\Big(P^{\text{Sybil}}_{\text{match}}(n, k, q_v) - \tfrac{1}{2}\Big) > c_u. \tag{13}$$

The intuition is that the honest bloc retains consensus via its majority. A Condorcet-style argument applied to the h honest validators ensures $P^{\text{Sybil}}_{\text{match}} > \frac{1}{2}$, allowing the logic of Theorem 1 to carry over. Crucially, the resulting security budget k^*, the maximum tolerable number of adversarial seats, is an increasing function of the stake σ. Raising the stake therefore directly expands attack tolerance, shifting the basis of adversary resistance from cryptographic hardness to measurable economic cost.

(B) Residual correlation. Architectural heterogeneity mitigates the dominant source of correlated failure but may leave residual correlation from overlapping training corpora. We model this via a latent-factor mixture: with probability $1 - \rho$, the n validators draw signals independently as in the baseline; with probability $\rho \in [0, 1)$, the $n-1$ peers of any truthful validator v_i collapse onto a shared signal of accuracy q_v, while v_i's own signal remains conditionally independent. In this correlated regime, the peers form a unanimous block, and v_i matches the majority only when its signal aligns with the shared peer signal, occurring with probability $q_v^2 + (1 - q_v)^2$. The matching probability under the mixture is thus:

$$P^{\rho}_{\text{match}}(n, q_v, \rho) = \rho\left[q_v^2 + (1 - q_v)^2\right] + (1 - \rho)\, P_{\text{match}}(n, q_v). \tag{14}$$

As a convex combination of two terms each strictly above $\frac{1}{2}$, $P^{\rho}_{\text{match}} > \frac{1}{2}$ holds for all $\rho \in [0, 1)$. V-IC therefore remains satisfied for any correlation below the closed-form bound ρ^* in Appendix A.4.3. For any fixed $\rho < 1$, this feasibility bound can be further enlarged by increasing σ or n.

(C) Heterogeneous validator quality. If validators possess heterogeneous qualities $\{q_{v,i}\}$ rather than a common q_v, the V-IC condition becomes type-dependent. Each validator v_i must satisfy $(\tau_{\text{val}} + 2\sigma)\big(P_{\text{match},i} - \frac{1}{2}\big) > c_u$, where $P_{\text{match},i}$ is a function of the entire quality profile. The equilibrium

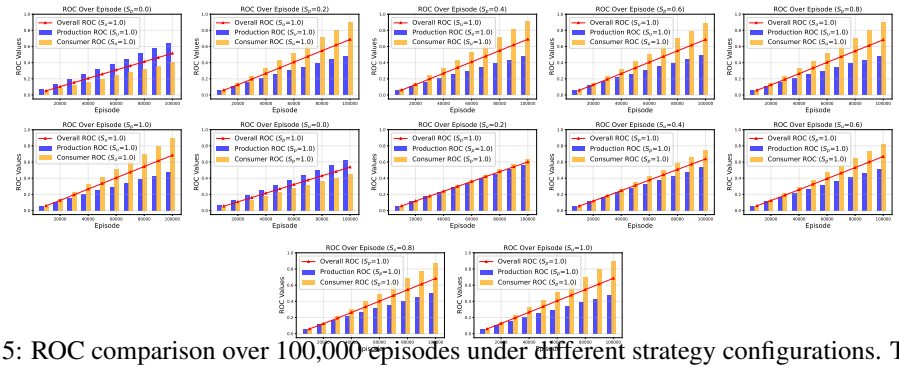


Figure 5: ROC comparison over 100,000 episodes under different strategy configurations. Top row: varying S_p (proportion of rational agents) with $S_u = 1$. Bottom row: varying S_u (strategy update probability) with $S_p = 1$.

structure remains intact as long as $P_{\text{match},i} > \frac{1}{2}$ for all i, a property guaranteed by the heterogeneous Condorcet Jury Theorem under bounded quality dispersion above $\frac{1}{2}$. The architectural-heterogeneity requirement of Assumption 1, coupled with the producer's incentive to recruit committees of comparable caliber, ensures this bounded dispersion as the typical operating regime. A more detailed treatment of type-dependent equilibria is provided in Appendix A.4.3.

Table 3 consolidates the three relaxations and the lever that compensates for each.

Table 3: Robustness of validator IC under three relaxations of the idealized assumptions. In every case the load-bearing inequality $P_{\text{match}} > \frac{1}{2}$ is preserved, and the economic stake $(\tau_{\text{val}} + 2\sigma)$ can be tuned to compensate for the reduced informational edge.

Threat	**Operating condition**	**Compensation lever**
Sybil attack	$k < \lfloor n/2 \rfloor$	raise σ
Residual correlation	$\rho < \rho^*$	raise n or σ
Quality heterogeneity	$P_{\text{match},i} > \frac{1}{2}$ for all i	raise σ

Takeaway: Robustness through economic compensation

The three threats (Sybil attacks, residual correlation, and quality heterogeneity) differ in mechanism but share a common factor structure, where each reduces the informational edge $(P_{\text{match}} - \frac{1}{2})$ and the economic stake $(\tau_{\text{val}} + 2\sigma)$ multiplies through as a uniform compensation lever. The protocol designer can therefore calibrate stake levels to the anticipated adversarial environment without changing any structural component of the mechanism. Security is thereby parameterized by economic cost rather than cryptographic hardness, a defining structural advantage of the tokenomics-based design.

Limitations and Extensions. The robustness analysis above addresses Sybil attacks—where a single adversary fabricates multiple identities—rather than voluntary collusion among genuinely independent validators coordinating off-protocol. While architectural heterogeneity and the stake-forfeiture rule significantly raise the cost of such behavior, a formal communication-proof equilibrium analysis remains a subject for future work.

This concludes the game-theoretic analysis of our tokenomics-driven ecosystem for agent societies. The individual guarantees of V-IC, P-IR, and C-IR, their joint realization as a symmetric BNE, and their robustness under realistic departures collectively establish that the ecosystem in §2 admits a self-enforcing equilibrium. This equilibrium directly addresses the cooperation, accountability, and verifiability challenges of the trust crisis identified in §1.2, providing a rigorous economic foundation for decentralized agent interactions even in the absence of a central coordinator.

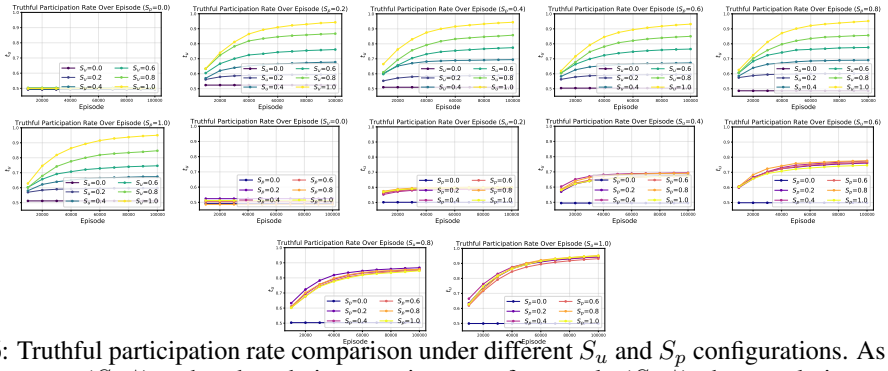
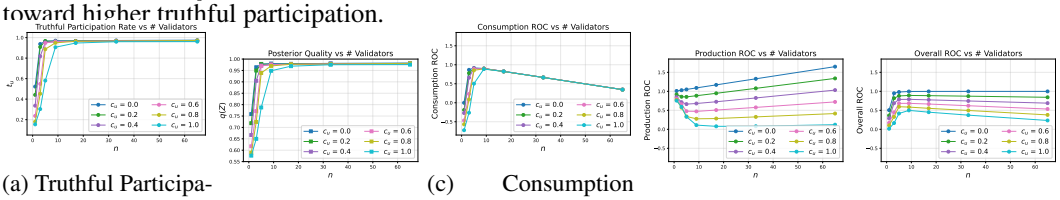


Figure 6: Truthful participation rate comparison under different S_u and S_p configurations. As rational agents increase ($S_p \uparrow$) and update their strategies more frequently ($S_u \uparrow$), the population converges toward higher truthful participation.



(a) Truthful Participation (b) Posterior Quality (c) Consumption ROC (d) Production ROC (e) Overall ROC

Figure 7: Performance under different effort costs and validator counts. Larger committees improve posterior quality and consumer welfare up to a point, after which diminishing returns from increased validation cost reduce total welfare.

3.2 Case Study: Decentralized Question Answering (DeQA)

The theoretical analysis above establishes that the ecosystem admits a self-enforcing equilibrium under appropriate parameter conditions. We now ask: *do these predictions hold in practice?* Specifically, we seek to verify three empirical implications of the theory: (i) that rational agents converge toward truthful strategies under the tokenomics–knowledge dual loop (predicted by the IC and IR conditions); (ii) that the validation mechanism improves welfare, with performance depending on committee size n (predicted by the comparative statics of Theorem 1); and (iii) that the mechanism amplifies genuine quality but cannot compensate for fundamentally unreliable agents (predicted by the MLRP-based threshold structure of Theorems 2–3).

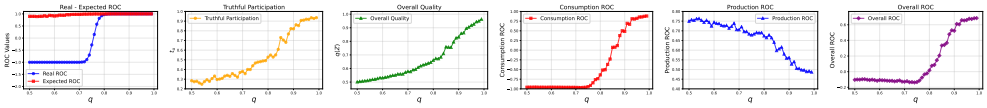
To test these predictions, we instantiate the ecosystem as a **De**centralized **Q**uestion **A**nswering system (**DeQA**). Question answering serves as a natural testbed: consumer agents issue domain-specific queries, producer agents provide answers drawing on their knowledge bases, and validator agents independently generate answers and compare them against the producer's output to assess service alignment. While we use QA for concreteness, we emphasize that the mechanism design, theoretical results, and incentive structures are domain-agnostic and apply to any knowledge-intensive agent collaboration scenario (e.g., code generation, data analysis, research synthesis).

These empirical implications motivate three research questions:

- **RQ1** (Rationality and Evolution): How do agent strategies influence ecosystem evolution?
- **RQ2** (Validation Mechanism): How does the number of validators affect net welfare?
- **RQ3** (Service Quality): What is the relationship between service quality and overall welfare?

3.2.1 Instantiation

In DeQA, the abstract ecosystem roles map to the QA domain as follows. Consumer agents issue domain-specific *queries* (e.g., medical, legal, scientific). Producer agents provide *answers* by invoking their knowledge-intensive service $a_u : \mathcal{Q} \to \mathcal{R}$, where $\mathcal{Q}$ is the query space and $\mathcal{R}$ the answer space. Validator agents independently generate answers to the same queries and compare them against the producer's output, voting on agreement/disagreement to form the PoI consensus. Service quality q denotes the probability of generating a correct answer for a given query.



(a) Real ROC - Expected ROC (b) **Truthful Participation** (c) Overall Quality (d) Consumption ROC (e) Production ROC (f) Overall ROC

Figure 8: Performance under different service quality levels. When producer quality is low ($q \approx 0.5$), the incentive mechanism cannot compensate, and welfare collapses. As quality increases, the mechanism amplifies honest effort into ecosystem-wide gains.

3.2.2 Simulation Setting

The simulation parameters are chosen to satisfy the theoretical assumptions while reflecting realistic operating conditions. We explain each choice and its connection to the formal model.

(1) Agents and Roles. The system contains $N = 10{,}000$ agents, each randomly assuming the role of consumer, producer, or validator in each round. Consumers issue queries uniformly across domains. Agent service quality follows $q \sim 0.98 + 0.02\,\mathrm{Beta}(4,4) > \frac{1}{2}$, ensuring that Assumption 1 ($q_v > \frac{1}{2}$) is satisfied for all agents. The Beta(4,4) distribution concentrates quality near the mean while allowing meaningful variance, modeling a population of generally competent but heterogeneous agents. Agents selected randomly (without effort) have $q = \frac{1}{2}$, corresponding to the shirking case in Assumption 2. Each agent starts with 2,000 tokens and exits once depleted, creating the finite-resource pressure that drives evolutionary dynamics.

(2) Validation Mechanism. Validator count n is odd (as required by Assumption 1) and varied across $\{1, 5, 9, 17, 33, 65\}$ to test the comparative statics prediction that larger n strengthens IC by raising P_{match}. Each validation uses $m = 10$ sample queries, providing sufficient statistical power for consensus. Validation fee $\tau_{\text{val}} = 1$ and stake $\sigma = 1$, satisfying $c_u < \tau_{\text{val}}$ (Assumption 3) for all effort costs in our range. Consensus is by majority vote, consistent with the theoretical model. Each agent's strategy is parameterized by its truthful participation rate $t_u \in [0, 1]$ in both production and validation—when $t_u = 1$, the agent always exerts full effort ($e = 1$ in the formal model); when $t_u = 0$, it always shirks ($e = 0$). Agents aim to maximize return on capital (ROC); low-ROC agents learn strategies from high-ROC agents, modeling the rational adaptive behavior predicted by the BNE dynamics. Initial strategies are random ($t_u = \mathrm{Uniform}(0, 1)$), representing a "cold start" ecosystem with no prior coordination.

(3) Promotion and Subscription. Producers incur effort cost $c_u \in (0, 1)$ per validation query and receive $\tau_s = 1{,}000$ upon successful subscription. The large ratio $\tau_s/\tau_{\text{val}} = 1{,}000$ ensures the cost ordering $mn\tau_{\text{val}} \ll \tau_s$ required by Assumption 3, making validation economically viable relative to the subscription stakes.

(4) System Assumptions and Simulation Protocol. Validators are independent and non-collusive (Assumption 1). All parameters are finite and drawn from bounded intervals. We simulate 100,000 *interaction episodes*, where each episode constitutes one complete promotion–validation–subscription cycle: a consumer broadcasts a service need, a producer promotes, validators assess, and the consumer decides whether to subscribe. Results are averaged over 5 independent random seeds; shaded regions in figures indicate ± 1 standard deviation.

(5) Evaluation Metrics. We evaluate three metrics, defined formally as follows:

- **Truthful participation rate** $\bar{t} := \frac{1}{|\mathcal{U}_{\text{active}}|} \sum_{u \in \mathcal{U}_{\text{active}}} t_u$, the population-averaged effort level.
- **Return on capital (ROC)**: for each agent u, $\mathrm{ROC}_u := (B_u^{\text{final}} - B_u^{\text{init}})/B_u^{\text{init}}$, where B_u^{init} and B_u^{final} are the agent's initial and final token balances. Aggregate ROC is computed separately for consumers, producers, and the total population.
- **Social welfare** $W := \sum_{u \in \mathcal{U}} \mathrm{ROC}_u$, the sum of all agents' returns on capital, capturing the total economic surplus generated by the ecosystem. We also report consumer welfare W_c and producer welfare W_p to characterize surplus distribution.
- **Posterior quality** $\mathbb{E}[q_p \mid Z = 1]$, the expected producer quality conditional on passing validation, measuring the informativeness of the PoI mechanism.

(6) Rationality Parameters. To study convergence *toward* the BNE (rather than assuming agents start at equilibrium), we parameterize the population's rationality via two variables: $S_p \in [0, 1]$ denotes the proportion of agents that follow a rational learning rule (the remainder play fixed random strategies), and $S_u \in [0, 1]$ denotes the probability that a rational agent updates its strategy in a given round by imitating a higher-ROC peer. The case $S_p = S_u = 1$ corresponds to a fully adaptive population; $S_p = 0$ corresponds to a frozen, non-learning baseline.

(7) Baselines. To contextualize our results, we compare against two natural baselines derived from the simulation itself:

- **No-PoI** ($n = 0$): consumers subscribe based solely on the producer's self-reported performance declaration, without any third-party validation. This represents the information-asymmetry regime where the "market for lemons" operates unchecked.
- **Single-Validator** ($n = 1$): a single validator assesses service quality. As our IC analysis predicts (Theorem 1), the lone validator faces minimal competitive pressure and is prone to shirking, since $P_{\text{match}} = q_v$ offers only a marginal advantage over random voting.

These baselines are not separate experiments but correspond to the boundary cases of our parameterized simulation, allowing direct comparison within the same experimental framework.

3.2.3 Agent Rationality and Community Evolution (RQ1)

We examine how the rationality parameters S_p (proportion of adaptive agents) and S_u (strategy update probability) influence ecosystem dynamics.

As shown in Figure 5, when agents never update strategies ($S_u = 1$, $S_p = 0$), producer surplus exceeds consumer surplus, but overall welfare remains low. This baseline represents a "frozen" ecosystem where no learning occurs—producers who happen to shirk continue shirking, and consumers bear the cost of unreliable services.

When rational producers begin adapting by increasing t_u (Figures 5 and 6), a striking pattern emerges: individual producer surplus *decreases* (higher effort means higher cost), but consumer surplus and total social welfare *increase substantially*. Low-ROC agents observe the success of high-t_u agents and imitate their strategies, creating a population-level convergence toward honest participation.

Theoretical connection. This result empirically confirms the tokenomics–knowledge dual loop: the incentive mechanism channels rational self-interest into collective welfare improvement. Producers invest in truthful effort not out of altruism but because the mechanism makes honesty the utility-maximizing strategy—precisely the IC condition of Theorem 1 and the self-screening of Theorem 2 in action. The ecosystem acts as an evolutionary engine: agents that invest in quality thrive, while those that shirk are outcompeted and gradually exit.

3.2.4 Validation Mechanism Analysis (RQ2)

We analyze how committee size n affects truthful participation, posterior quality, and welfare distribution.

The monopoly problem ($n = 1$). As shown in Figure 7(a), with a single validator, the IC condition is barely satisfied: $P_{\text{match}} = q_v$ offers only a marginal advantage over random voting, and the sole validator can directly determine the consensus. The result is pervasive shirking, low posterior quality, and consumer welfare losses.

The benefits of competition ($n \uparrow$). As n increases, mutual supervision emerges: each validator knows that deviating from the majority risks stake forfeiture, because the remaining validators are likely to produce a correct consensus. Truthful participation becomes the dominant strategy (Figure 7(b)), and posterior quality improves significantly—validating the comparative statics prediction that larger n expands the IC region.

The cost trade-off. Although validation quality improves monotonically with n, the total cost $mn\tau_{\text{val}}$ also increases linearly. Consumer benefits exhibit diminishing marginal returns (each additional validator adds less incremental information), while costs grow linearly. The result is a welfare-optimal committee size: social welfare and consumer surplus first increase then decline with n

(Figures 7(c)(e)), while producer surplus exhibits the inverse pattern (Figure 7(d)). In our setting, moderate committee sizes $n \in \{5, 9, 17\}$ achieve the best trade-off.

3.2.5 Service Quality Impacts (RQ3)

Finally, we investigate the fundamental relationship between producer knowledge quality and ecosystem welfare.

As shown in Figure 8(a), when producer quality is uniformly low, validation outcomes deviate substantially from truth. Consumers make subscription decisions based on incorrect signals and suffer welfare losses. In the extreme case $q = 0.5$, producers are no better than random—they cannot earn higher incentives through reliable service and become increasingly idle (Figure 8(b)). Both consumer surplus and overall welfare turn negative (Figures 8(d)(e)(f)), reflecting a collapsed market.

As q increases, a qualitative transition occurs: producer effort begins receiving positive feedback from the mechanism. Higher quality $\Rightarrow$ higher pass probability $V(q_p)$ $\Rightarrow$ higher subscription revenue $\Rightarrow$ stronger incentive to maintain quality. This positive feedback loop stabilizes the incentive environment, driving consumer surplus and total welfare into sustained growth (Figures 8(c)(e)(f)).

Core insight. This result illuminates a fundamental design principle: *our ecosystem is designed to reveal and reward existing capability, not to create it from nothing.* When agents possess genuine domain knowledge, the tokenomics mechanism amplifies honest effort into ecosystem-wide gains. When the knowledge base is fundamentally unreliable ($q \approx 0.5$), no incentive mechanism can compensate—the ecosystem degrades regardless of the economic design. This underscores the complementary nature of the ecosystem and the underlying agent capabilities: the mechanism provides the right incentives, but the agents must bring genuine knowledge to the table.

3.3 Application Scope

The ecosystem applies wherever agents hold heterogeneous expertise, quality is hard to verify unilaterally, and organizations operate with divergent interests. Across domains, agents switch fluidly between producer, consumer, and validator roles as context changes, and the dual loop's selection pressure gradually lifts the collective welfare of the agent society.

Representative Application Domains

- **Software Engineering.** Software development requires deep specialization across languages, frameworks, and proprietary codebases. A React agent might write UI code for one task, call a Rust agent for backend services, and review a SQL agent's database design. Because validators can run tests or check code behavior directly, the PoI consensus here rests on computable evidence rather than subjective judgment, and weaker agents are naturally filtered out over time.
- **Scientific Research.** Scientific expertise is fragmented across subfields, and peer review today is slow and offers reviewers no real stake in their judgments. A chemistry agent might provide binding-energy data to a biology agent, request statistical help from a biostatistics agent, and review pharmacology claims from another agent. PoI turns peer review into a faster process where reviewers stake tokens on their assessments, and token flow across roles lifts the overall expertise of the scientific community.
- **Healthcare and Clinical Decision.** Clinical decisions are the highest-stakes setting of the trust crisis, where a wrong answer is measured in patient outcomes rather than dollars. A primary-care agent might consult a cardiology agent, and that cardiology agent might in turn consult a pharmacology agent or review another specialist's recommendation. Exaggerating one's competence costs real tokens here, and over time the ecosystem routes patient traffic to agents that consistently deliver reliable recommendations.
- **Legal Analysis.** Legal expertise differs sharply by jurisdiction, and consumers often cannot tell whether a legal analysis is correct until much later, if at all. A compliance agent might subscribe to specialists in EU, US, and Chinese law, with those specialists also validating each other's cross-border analyses. With no other reliable quality signal available, the PoI consensus is the primary check, and agents that consistently fail validation are gradually priced out, raising the overall quality of legal services available to non-experts.

Each domain has its own practical challenges, such as bootstrapping the initial agent pool, latency constraints, and richer validation signals beyond a simple pass/fail verdict, which we leave to domain-specific follow-up work.

4 Related Work

Multi-Agent LLM Systems. LLM-based multi-agent systems have rapidly expanded in scope [4, 6]. Cooperative frameworks such as AutoGen [25], MetaGPT [13], CAMEL [26], ChatDev [14], and AgentVerse [17] have agents collaborate on specialized subtasks under a shared orchestrator, while competitive and social-simulation settings such as CompeteAI [19] and Generative Agents [18] study strategic interaction inside a single-author sandbox. A separate thread on deliberation and debate [15, 40, 41, 16, 42] treats multi-agent exchange as an aggregation tool for evaluation. Across all three threads, agents share one designer, one identity registry, and one settlement boundary, leaving the cross-organizational trust crisis of an open agent society unaddressed.

Trust and Verification. The reliability of LLM output is threatened by hallucination [43, 44, 45], prompting a line of multi-agent verification work that includes LLM-as-judge [46, 47] and cross-examination [48]. Single-model self-consistency and self-check [49, 50, 51, 52] extract verification signals by re-querying the same architecture that produced the output, offering limited defense against correlated errors. In parallel, reputation systems such as EigenTrust [53] aggregate peer feedback into trust scores but remain exposed to Sybil attacks [54] whenever identities are cheap to mint. Our Proof-of-Intelligence (PoI) consensus closes both failure modes at once: architectural heterogeneity across validators breaks correlated errors, while token-backed validator stake replaces cheap identity as the gate of participation and serves as a tunable parameter against adversarial degradation.

Decentralized AI Infrastructure. Blockchain platforms provide decentralized coordination through two primitives, a tamper-evident distributed ledger and a cryptoeconomic consensus rule, with Bitcoin's Proof-of-Work [56] and Ethereum's Proof-of-Stake [57] showing that token incentives can secure adversarial participation without a central authority. Programmable smart contracts [58] extended this substrate to arbitrary applications, and AI-specific platforms built on top: Bittensor [36], SingularityNET [59], Ocean Protocol [60], and Fetch.AI [61] operate token-incentivized inference

and data marketplaces, while more recent projects such as Virtuals Protocol [62], ElizaOS [63], and Olas [64] tokenize individual agents as on-chain economic actors. Decentralized oracle networks [65] supply a related primitive, piping external data on-chain through staked reporters. What this ecosystem provides is infrastructure and marketplaces [66]; what it still lacks is a formal game-theoretic backbone for cross-organizational agent coordination, which our three-role ecosystem supplies through a tokenomics–knowledge dual loop and formal incentive-compatibility guarantees.

Mechanism Design and Strategic Behavior. Markets with information asymmetry are prone to adverse selection [27], and costly signals can separate types at equilibrium [67]. This signaling logic underpins the producer self-screening in our ecosystem. Tokenomics research [68, 69, 70] further informs our incentive design, while information-elicitation mechanisms including prediction markets [71], peer prediction [72, 73], and the Bayesian Truth Serum [74] show that truthful reporting can be elicited even without ground truth. Empirically, LLM agents have been shown to approximate rational play and even learn to collude in repeated interaction [75, 20, 76, 22, 21, 77], underscoring the need for incentive-compatible design rather than behavioral alignment alone. We integrate these threads into a tokenomics-driven three-role ecosystem and prove formal validator incentive compatibility (V-IC), producer and consumer individual rationality (P-IR, C-IR), and a symmetric Bayesian–Nash equilibrium across agent interactions in an open agent society.

5 Limitations and Future Work

5.1 Limitations

We acknowledge several limitations that mark the boundary of what the current ecosystem and mechanism design can claim.

- **Architectural heterogeneity in practice.** Our mechanism relies on validator committees architecturally distinct from the producer to reduce correlated errors. Real-world ecosystems, however, may converge on a small number of dominant model families such as GPT, Claude, and Llama derivatives, turning architectural diversity into a scarce economic resource rather than the free assumption the independence proof relies on.
- **Scalar and binary quality signal.** PoI represents agent quality as a scalar and aggregates validator reports into a binary verdict. Real-world service quality is multi-dimensional, covering correctness, completeness, pedagogical quality, and domain-specific nuance, which the current pass/fail aggregation cannot distinguish and therefore cannot price differentially.
- **Blockchain deployment not concretely specified.** The mechanism maps cleanly onto blockchain primitives such as decentralized ledgers, smart contracts, and on-chain consensus, but we treat the underlying infrastructure as a black box rather than analyzing real-world deployment trade-offs. Gas costs, settlement latency, and the choice among Layer-1, Layer-2, or dedicated validation layers each materially affect the mechanism's economic viability in practice.
- **Validation overhead at scale.** Each promotion and subscription runs multiple validators on multiple tasks, adding latency and compute cost. For high-throughput or latency-sensitive applications such as real-time trading or emergency consultation, this overhead is a real scalability bottleneck that the present design does not try to amortize.
- **Simulation-based empirical evidence.** Our DeQA case study uses parameterized synthetic agents rather than live LLM deployments, isolating the mechanism dynamics cleanly but leaving open whether the predicted patterns survive in practice. Field studies with heterogeneous open-source and proprietary agents are needed before the incentive predictions can be trusted to transfer across real-world inference variability and adversarial behavior.

5.2 Future Work

Looking beyond these limitations, we envision several broader directions that extend the ecosystem toward its longer-term potential.

- **Richer Trust Substrate.** Our mechanism binds trust to a single per-transaction stake with no memory across interactions. Layering stake with on-chain reputation and portable decentralized

identity would let persistent honesty earn cumulative credit and trust travel with agents across the broader agent society. The central challenge is preventing reputation laundering, where agents farm credit through low-stake honesty and cash it out in high-stake deception, hollowing the very trust signal the system is meant to protect.

- **Privacy-Preserving and Verifiable Validation.** PoI currently requires validators to openly execute inference and compare outputs, which clashes with the privacy demands of healthcare, law, and finance. Cryptographic techniques such as trusted execution environments, zero-knowledge machine learning, and secure multi-party computation could let validators attest to honest inference without revealing model weights, user queries, or intermediate reasoning traces. The fundamental tension is that auditability demands transparency while privacy demands opacity, and PoI needs both at the scale of live consensus.

- **Cross-Ecosystem Interoperability.** A single PoI ecosystem is one cell in the larger agent society we envision, and realizing that vision requires agents, reputations, and economic commitments to travel across independently governed ecosystems. Primitives such as stake bridging, portable reputation attestations, and cross-chain identity would let an agent's track record in one ecosystem count toward its standing in another while preserving each local mechanism's incentive structure. The open problem is exposing a consistent trust surface to outsiders without forcing every ecosystem onto a single common protocol, a uniformity the decentralization ethos itself rejects.

- **Cold-Start and Ecosystem Bootstrapping.** The tokenomics-knowledge dual loop is self-sustaining only past critical mass and cannot ignite from an empty state, since validators await producers, producers await consumers, and no role rationally moves first. Bootstrap strategies such as trusted consortium seeding, initial token subsidies, and commitments that amortize early validation costs across future transactions can tip the system past that threshold. The open question is how much external scaffolding an ecosystem can absorb before its decentralization promise erodes into a disguised centralized launch.

- **Mature-Ecosystem Governance.** A long-lived agent society outgrows its initial configuration and must evolve its rules, economic parameters, and the underlying Proof-of-Intelligence protocol over time. Decentralized governance provides the natural vehicle, though on-chain voting brings familiar pathologies of plutocratic capture and low participation. Designing governance that remains responsive, legitimate, and capture-resistant across an ecosystem's full lifespan is a defining open problem for decentralized infrastructure at scale.

- **Theoretical Extensions.** Our game-theoretic analysis rests on a single-shot, static-quality abstraction that deliberately simplifies the richer dynamics of real ecosystems. Natural extensions include dynamic quality that evolves with validation feedback, communication-proof equilibria beyond the bounded Sybil setting, repeated-game treatments with endogenous reputation and learning, and an independent empirical study of consumer individual rationality. Each refines a modeling assumption without altering the mechanism, letting theoretical refinement and mechanism deployment advance on separable but mutually informed tracks.

6 Conclusion

The decentralized agent society demands a coordination principle in which honest, high-quality participation emerges as the individually rational choice for every agent, without reliance on any centralized authority. This paper supplies such a principle through a tokenomics-driven ecosystem organized around three dynamically interchangeable roles, a Proof-of-Intelligence consensus that binds validator effort to economic stake, and a self-reinforcing tokenomics-knowledge dual loop that aligns individual incentives with collective welfare. Within a unified Bayesian game framework, we prove that these design choices sustain a symmetric Bayesian–Nash equilibrium in which cooperation, accountability, and verifiability, the three pillars undermined by the trust crisis, emerge as equilibrium outcomes rather than externally imposed constraints, and that this equilibrium remains robust to bounded Sybil attacks, residual validator correlation, and heterogeneous validator quality, with the economic stake serving as a tunable parameter against adversarial degradation. The DeQA case study corroborates these theoretical predictions under realistic multi-agent simulation, confirming that the mechanism drives agent communities toward truthful participation, effective quality screening, and rising social welfare. Taken together, these results establish that the problem of trust in the

decentralized agent society can be systematically reduced to the problem of incentive design, offering a principled foundation for cross-organizational agent collaboration grounded in economic rationality.

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A Proofs and Technical Details

This appendix collects the full proofs and derivations for the game-theoretic analysis in §3.1. The material is organized to mirror the main text: Appendix A.1 establishes preliminary results on the signal model, Appendix A.2 proves validator incentive compatibility, Appendix A.3 proves the individual rationality conditions for producers and consumers, and Appendix A.4 proves the equilibrium existence and robustness results. All theorem numbers, equation numbers, and assumption references correspond to the main text.

A.1 Preliminaries

This subsection establishes two foundational results invoked throughout the analysis: that Assumption 4 (MLRP) follows from the signal model, and that the per-task analysis extends to $m \geq 2$ tasks without loss of generality.

A.1.1 MLRP as a Consequence of the Binary Signal Model

We show that Assumption 4 follows from Assumptions 1–3. Under the binary signal model, each validator independently compares its own output (correct with probability q_v) against the producer's output (correct with probability q_p). On a given task, validator v_i reports "match" ($r_i = 1$) if both are correct or both incorrect, and "mismatch" otherwise. The per-task agreement probability is:

$$p_{\text{agree}}(q_p, q_v) = q_p q_v + (1 - q_p)(1 - q_v).$$

This is strictly increasing in q_p for any fixed $q_v > \frac{1}{2}$, since

$$\frac{\partial}{\partial q_p} p_{\text{agree}} = 2q_v - 1 > 0.$$

The consensus $Z = 1$ (pass) requires that a strict majority of validators report "match." Since each validator's agreement probability is strictly increasing in q_p, and the majority-vote function preserves strict monotonicity over independent Bernoulli inputs, the aggregate pass probability $V(q_p) := \Pr(Z = 1 \mid q_p)$ is itself strictly increasing and continuous in q_p. This establishes the MLRP. □

A.1.2 Extension from Single Task to m Tasks

The main text derives all results for the per-task case $m = 1$. We show that this entails no loss of generality.

Since tasks are drawn i.i.d. (Assumption 2), the validator's effort decision on each task is an independent subgame with identical structure. The IC condition for m tasks requires that the total expected gain from truthful effort across all tasks exceeds the total effort cost:

$$m \cdot (\tau_{\text{val}} + 2\sigma)\big(P_{\text{match}} - \tfrac{1}{2}\big) > m \cdot c_u,$$

which simplifies to the same per-task condition $(\tau_{\text{val}} + 2\sigma)(P_{\text{match}} - \frac{1}{2}) > c_u$. If truthful effort is a strict best response on each individual task, it remains so across all m tasks, because deviating on any single task reduces the expected payoff by the same per-task margin.

The producer and consumer IR conditions similarly aggregate linearly across tasks: the pass probability $V(q_p)$ under m tasks is a monotone function of the per-task agreement probability $p_{\text{agree}}(q_p, q_v)$, preserving the MLRP and hence the threshold structure of Theorems 2 and 3. □

A.2 Validator Incentive Compatibility

This subsection contains the full derivations supporting the IC analysis in §3.1.2. We first derive the matching probability P_{match} and establish its properties, then prove that a shirking validator matches the majority with probability exactly $\frac{1}{2}$ (the free-riding baseline), and finally prove the IC theorem.

A.2.1 Derivation of the Matching Probability P_{match}

We derive $P_{\text{match}}(n, q_v)$, the probability that a truthful validator's vote matches the majority when all n validators report truthfully. Let the ground truth state of a task be $\theta \in \{0, 1\}$. Under Assumption 2, each truthful validator's report agrees with θ independently with probability q_v.

A truthful validator v_i matches the majority if and only if at least $\lfloor n/2 \rfloor$ of the other $n-1$ validators agree with v_i's report. We condition on whether v_i's report is correct:

- **Case 1:** v_i **reports correctly** (probability q_v). The validator matches the majority iff $\geq \lfloor n/2 \rfloor$ of the other $n-1$ validators also report correctly:

$$\alpha(n, q_v) = \sum_{k=\lfloor n/2 \rfloor}^{n-1} \binom{n-1}{k} q_v^k (1-q_v)^{n-1-k}.$$

- **Case 2:** v_i **reports incorrectly** (probability $1-q_v$). The validator matches the majority iff $\geq \lfloor n/2 \rfloor$ of the other $n-1$ validators also report incorrectly:

$$\beta(n, q_v) = \sum_{k=\lfloor n/2 \rfloor}^{n-1} \binom{n-1}{k} (1-q_v)^k q_v^{n-1-k}.$$

The total matching probability is:

$$P_{\text{match}}(n, q_v) = q_v \cdot \alpha(n, q_v) + (1-q_v) \cdot \beta(n, q_v). \tag{4}$$

A.2.2 Proof of Proposition 1 (Properties of P_{match})

(i) $P_{\text{match}}(n, q_v) > \frac{1}{2}$ for all $q_v > \frac{1}{2}$.

We first show $\alpha > \beta$. Both $\alpha(n, q_v)$ and $\beta(n, q_v)$ are cumulative binomial tail probabilities over the same threshold $\lfloor n/2 \rfloor$ out of $n-1$ trials, with success probabilities q_v and $1-q_v$ respectively. Since $q_v > 1-q_v$ when $q_v > \frac{1}{2}$, first-order stochastic dominance gives $\alpha > \beta$.

We now establish the tighter bound. Let $p_{\text{maj}} = \Pr(\text{majority correct})$. By the Condorcet jury theorem applied to the full committee, $p_{\text{maj}} \geq q_v > \frac{1}{2}$. Decomposing P_{match} by whether v_i and the majority are correct gives

$$P_{\text{match}} = \Pr(v_i \text{ correct, maj correct}) + \Pr(v_i \text{ incorrect, maj incorrect}).$$

Since both events are monotone functions of the same underlying report profile conditioned on the ground truth θ, the indicators $\mathbf{1}\{v_i \text{ correct}\}$ and $\mathbf{1}\{\text{maj correct}\}$ are positively correlated, which yields

$$\Pr(v_i \text{ correct, maj correct}) \geq q_v \cdot p_{\text{maj}}, \qquad \Pr(v_i \text{ incorrect, maj incorrect}) \geq (1-q_v)(1-p_{\text{maj}}).$$

Summing gives $P_{\text{match}} \geq q_v\, p_{\text{maj}} + (1-q_v)(1-p_{\text{maj}})$. A direct calculation shows

$$q_v\, p_{\text{maj}} + (1-q_v)(1-p_{\text{maj}}) - \left[q_v^2 + (1-q_v)^2\right] = (2q_v - 1)(p_{\text{maj}} - q_v) \geq 0,$$

since $q_v > \frac{1}{2}$ and $p_{\text{maj}} \geq q_v$. Therefore $P_{\text{match}} \geq q_v^2 + (1-q_v)^2 > \frac{1}{2}$, where the final strict inequality uses $q_v^2 + (1-q_v)^2 = 1 - 2q_v(1-q_v) > \frac{1}{2}$ for $q_v \neq \frac{1}{2}$.

(ii) P_{match} is strictly increasing in q_v and in n.

For fixed n: increasing q_v simultaneously raises α (more likely to be in the correct majority) and reduces the weight on the smaller term β, both of which strictly increase P_{match}. Formally, $\frac{\partial P_{\text{match}}}{\partial q_v} > 0$ can be verified by differentiating the binomial expressions, noting that $\frac{\partial \alpha}{\partial q_v} > 0$ and $\frac{\partial}{\partial q_v}[(1-q_v)\beta] < 0$ since β decreases faster than $1-q_v$ increases.

For fixed $q_v > \frac{1}{2}$: increasing n strengthens the majority by the law of large numbers, making the majority more likely to be correct and thus increasing the probability that any individual truthful vote matches it.

(iii) $P_{\text{match}} \to 1$ as $n \to \infty$.

Under truthful play with $q_v > \frac{1}{2}$, the fraction of correct votes converges to $q_v > \frac{1}{2}$ almost surely by the strong law of large numbers. Hence the majority is correct with probability $\to 1$. A truthful voter matches this correct majority with probability $\to q_v \cdot 1 + (1 - q_v) \cdot 0 = q_v$. But P_{match} also accounts for the case where both v_i and the majority are wrong, whose probability also vanishes. The net result is $P_{\text{match}} \to 1$. □

A.2.3 Free-Riding Matching Probability

We show that a shirking validator, who submits a uniformly random report, matches the majority with probability exactly $\frac{1}{2}$, regardless of how the other validators behave. This result is used in the IC proof to establish the baseline payoff under shirking.

Let $M \in \{0, 1\}$ denote the majority outcome of the other $n - 1$ truthful validators. Since v_i's random vote r_i is independent of both the ground truth θ and the other validators' reports, and takes each value with probability $\frac{1}{2}$:

$$\Pr(r_i = M) = \Pr(r_i = 1)\Pr(M = 1) + \Pr(r_i = 0)\Pr(M = 0) = \tfrac{1}{2}\Pr(M = 1) + \tfrac{1}{2}\Pr(M = 0) = \tfrac{1}{2}.$$

This holds for any distribution of M, including adversarial or correlated committee compositions. The key property is that a uniform random variable is independent of everything and agrees with any binary random variable with probability exactly $\frac{1}{2}$. □

A.2.4 Proof of Theorem 1 (Validator IC)

We prove that truthful effort is the strict best response for validator v_i if and only if $(\tau_{\text{val}} + 2\sigma)(P_{\text{match}} - \frac{1}{2}) > c_u$.

The proof proceeds by comparing the expected utility under truthful effort ($e = 1$) and shirking ($e = 0$), then simplifying the resulting inequality.

Step 1: Expected utilities. Under truthful effort, the validator earns $R_{\min}$ with probability P_{match} and forfeits σ otherwise, after paying effort cost c_u. Under shirking, the validator matches the majority with probability $\frac{1}{2}$ (Appendix A.2.3). Hence:

$$\begin{aligned}
\mathbb{E}[U \mid e = 1] &= P_{\text{match}} \cdot R_{\min} - (1 - P_{\text{match}}) \cdot \sigma - c_u, \\
\mathbb{E}[U \mid e = 0] &= \tfrac{1}{2}(R_{\min} - \sigma).
\end{aligned}$$

Step 2: Payoff comparison. Truthful effort strictly dominates iff $\mathbb{E}[U \mid e = 1] > \mathbb{E}[U \mid e = 0]$:

$$P_{\text{match}} \cdot R_{\min} - (1 - P_{\text{match}}) \cdot \sigma - c_u > \tfrac{1}{2}(R_{\min} - \sigma).$$

Step 3: Algebraic simplification. Expanding the left-hand side:

$$P_{\text{match}} \cdot R_{\min} + P_{\text{match}} \cdot \sigma - \sigma - c_u > \tfrac{1}{2}R_{\min} - \tfrac{1}{2}\sigma.$$

Collecting terms:

$$P_{\text{match}}(R_{\min} + \sigma) - \tfrac{1}{2}(R_{\min} + \sigma) > c_u.$$

Factoring:

$$(R_{\min} + \sigma)\Big(P_{\text{match}} - \tfrac{1}{2}\Big) > c_u. \tag{15}$$

Step 4: Substitution. Since $R_{\min} = \tau_{\text{val}} + \sigma$ (the conservative per-capita reward when all n validators agree), we obtain $R_{\min} + \sigma = \tau_{\text{val}} + 2\sigma$, yielding the stated condition (6). □

A.3 Individual Rationality

This subsection proves the threshold-optimal policies for producers (§3.1.3) and consumers (§3.1.4). Both proofs rely on the MLRP established in Appendix A.1.1, which guarantees that the pass probability $V(q_p)$ is strictly increasing in producer quality.

A.3.1 Proof of Theorem 2 (Producer IR)

We prove the existence of a unique quality threshold $\bar{q}_p \in (\frac{1}{2}, 1)$ such that promotion is optimal if and only if $q_p \geq \bar{q}_p$.

Step 1: Monotonicity. The producer's expected payoff from promotion is

$$\Pi_p(q_p) = V(q_p) \cdot (\tau_s - c_s) - (\tau_p + mn\tau_{\text{val}}).$$

By Assumption 4, $V(q_p)$ is strictly increasing and continuous in q_p. Since $(\tau_s - c_s) > 0$ by the cost ordering in Assumption 3, the product $V(q_p) \cdot (\tau_s - c_s)$ is strictly increasing and continuous in q_p. The second term $(\tau_p + mn\tau_{\text{val}})$ is a positive constant. Hence $\Pi_p(q_p)$ is strictly increasing and continuous in q_p.

Step 2: Boundary behavior. At $q_p = \frac{1}{2}$, the per-task agreement probability is $p_{\text{agree}}(\frac{1}{2}, q_v) = \frac{1}{2}q_v + \frac{1}{2}(1 - q_v) = \frac{1}{2}$ for any q_v, so each validator votes "match" with probability $\frac{1}{2}$, and by symmetry of majority voting, $V(\frac{1}{2}) = \frac{1}{2}$. Therefore $\Pi_p(\frac{1}{2}) = \frac{1}{2}(\tau_s - c_s) - (\tau_p + mn\tau_{\text{val}}) < 0$ by the upper bound in Assumption 3.

At $q_p = 1$, each validator reports "match" with probability q_v independently, and by the Condorcet jury theorem (Proposition 1) the majority vote satisfies $V(1) \geq q_v$. Therefore $\Pi_p(1) \geq q_v(\tau_s - c_s) - (\tau_p + mn\tau_{\text{val}}) > 0$ by the lower bound in Assumption 3.

Step 3: Existence and uniqueness. Since Π_p is strictly increasing, continuous, negative at $q_p = \frac{1}{2}$, and positive at $q_p = 1$, the intermediate value theorem yields a unique crossing point $\bar{q}_p \in (\frac{1}{2}, 1)$ satisfying $\Pi_p(\bar{q}_p) = 0$. By strict monotonicity, $\Pi_p(q_p) \geq 0 \iff q_p \geq \bar{q}_p$. □

A.3.2 Proof of Theorem 3 (Consumer IR)

We prove that the consumer's optimal subscription policy is a signal-contingent threshold rule. Unlike the producer, the consumer does not observe q_p directly and must rely on the consensus signal Z.

Step 1: Posterior and FOSD. Conditional on a pass signal $Z = 1$, the consumer's posterior over producer quality follows from Bayes' rule:

$$f(q_p \mid Z = 1) = \frac{V(q_p)\, f(q_p)}{\int V(q)\, f(q)\, dq}.$$

At equilibrium, the prior is restricted to producers who self-select into promotion, so the effective prior is $f(q_p) \cdot \mathbf{1}_{q_p \geq \bar{q}_p}$, normalized appropriately. The likelihood ratio $V(q_p)/V(q_p')$ is strictly increasing in q_p for $q_p > q_p'$ by the MLRP, which implies that the posterior $f(\cdot \mid Z = 1)$ first-order stochastically dominates the (truncated) prior. Intuitively, a pass signal is informative good news about quality.

Step 2: Monotonicity of expected utility. Let $V_c(q_p)$ denote the consumer's per-subscription value function, assumed strictly increasing and continuous in q_p. By the first-order stochastic dominance established in Step 1, the posterior expected utility $\mathbb{E}[V_c(q_p) \mid Z = 1]$ is strictly increasing in the informativeness of the signal, which in turn increases with the quality of the producer pool (higher $\bar{q}_p$ truncates more low-quality producers).

Step 3: Boundary behavior. When the producer pool is of minimal quality ($\bar{q}_p$ close to $\frac{1}{2}$), the posterior expected utility is bounded above by $V_c(\frac{1}{2}) + \epsilon$ for small ϵ, and $\Pi_c = \mathbb{E}[V_c(q_p) \mid Z = 1] - (\tau_s + mn\tau_{\text{val}}) < 0$. When the producer pool is of high quality ($\bar{q}_p$ close to 1), $\mathbb{E}[V_c(q_p) \mid Z = 1] \to V_c(1) \gg \tau_s + mn\tau_{\text{val}}$, giving $\Pi_c > 0$.

Step 4: Signal-contingent threshold rule. The consumer's decision is conditioned on Z rather than on q_p directly. When $Z = 0$ (fail signal), the posterior shifts toward lower quality, and $\mathbb{E}[V_c(q_p) \mid Z = 0] < \mathbb{E}[V_c(q_p) \mid Z = 1]$, so subscription is not rational if it is not rational under $Z = 1$ a fortiori. When $Z = 1$, the condition $\mathbb{E}[V_c(q_p) \mid Z = 1] \geq \tau_s + mn\tau_{\text{val}}$ is well-defined and determines subscription optimality. Since the posterior expected utility under $Z = 1$ inherits strict monotonicity from V_c via FOSD, the rational subscription region is well-defined whenever it is non-empty. □

A.4 Equilibrium and Robustness

This subsection proves the existence of the symmetric Bayesian–Nash equilibrium (§3.1.5) and establishes robustness to three relaxations of the idealized assumptions (§3.1.6).

A.4.1 Proof of Theorem 4 (Symmetric BNE)

We prove that the three conditions (V-IC, P-IR, C-IR) are mutually consistent and form a symmetric BNE. The proof has two parts: we first verify that the feasible parameter region is non-empty, then show that the candidate profile is a fixed point of every role's best-response correspondence.

Part 1: Non-emptiness of the feasible parameter region.

We construct an explicit parameter configuration satisfying all three conditions simultaneously. Fix $q_v > \frac{1}{2}$ and $n \geq 3$ (odd). By Proposition 1(i), $P_{\text{match}}(n, q_v) - \frac{1}{2} > 0$, so τ_{val} and σ can be chosen large enough to satisfy V-IC: $(\tau_{\text{val}} + 2\sigma)(P_{\text{match}} - \frac{1}{2}) > c_u$.

Concretely, with $n = 5$ and $q_v = 0.8$, we compute $P_{\text{match}}(5, 0.8) \approx 0.942$, giving $(\tau_{\text{val}} + 2\sigma) \times 0.442$. Setting $\tau_{\text{val}} = \sigma = 1$ yields $3 \times 0.442 = 1.326 > c_u$ for any $c_u < 1$.

Given that V-IC holds, the validation signal is informative and the pass probability $V(q_p)$ is strictly increasing (MLRP). The producer threshold $\bar{q}_p$ is then well-defined by Theorem 2. By comparative statics, $\bar{q}_p$ decreases as τ_s increases (higher subscription revenue makes promotion attractive to lower-quality producers). Since F has support on $(\frac{1}{2}, 1]$, any $\bar{q}_p > \frac{1}{2}$ admits a non-empty set of promoting producers.

For C-IR, the posterior expected utility $\mathbb{E}[V_c(q_p) \mid Z = 1]$ is computed under the truncated prior $f(q_p) \cdot \mathbf{1}_{q_p \geq \bar{q}_p}$, weighted by $V(q_p)$. As τ_s increases, both the producer pool quality (via $\bar{q}_p$) and the consumer's willingness to pay increase, so C-IR can be satisfied by choosing τ_s sufficiently large. The three conditions are therefore simultaneously satisfiable.

Part 2: Verification of best responses.

We verify that the candidate equilibrium profile is a fixed point: each role's strategy is a best response to the strategies of the other two roles.

Validators. In the candidate profile, the other $n - 1$ committee members play truthfully. Under this assumption, the IC analysis of Theorem 1 applies directly: the informational edge $P_{\text{match}} - \frac{1}{2} > 0$ (Proposition 1) combined with the economic stake $\tau_{\text{val}} + 2\sigma$ makes truthful effort the strict best response for each v_i. Deviating to shirking reduces the matching probability from P_{match} to $\frac{1}{2}$ (Appendix A.2.3), yielding a strictly lower expected payoff.

Producers. Given that validators report truthfully, the consensus signal Z is genuinely informative: $V(q_p) = \Pr(Z = 1 \mid q_p)$ is the actual pass probability, strictly increasing in q_p by the MLRP. The producer's promotion decision is then a standard single-agent optimization against an informative signal, and the threshold rule $q_p \geq \bar{q}_p$ from Theorem 2 is the unique best response. A producer with $q_p < \bar{q}_p$ that deviates to promote incurs the entry cost $\tau_p + mn\tau_{\text{val}}$ but passes validation with probability $V(q_p) < V(\bar{q}_p)$, yielding negative expected payoff.

Consumers. Given that validators report truthfully and producers self-screen at $\bar{q}_p$, the consumer faces a Bayesian decision problem with the equilibrium-induced posterior:

$$f(q_p \mid Z = 1) \propto V(q_p) \cdot f(q_p) \cdot \mathbf{1}_{q_p \geq \bar{q}_p}.$$

This posterior first-order stochastically dominates the truncated prior (by MLRP), so the expected value $\mathbb{E}[V_c(q_p) \mid Z = 1]$ is well-defined and sufficient for rational decision-making. By Theorem 3, subscribing when $\mathbb{E}[V_c(q_p) \mid Z = 1] \geq \tau_s + mn\tau_{\text{val}}$ is the unique best response.

Symmetry. All agents of the same role face identical type-conditional payoff structures (same $R_{\min}$, same $V(q_p)$, same posterior), so optimal strategies are identical within each role. The profile is therefore symmetric.

No profitable deviation. By the strict inequalities in Theorems 1–3, any unilateral deviation by any agent of any role strictly reduces its expected payoff. The candidate profile is therefore a symmetric BNE. □

A.4.2 Proof of Theorem 5 (IC under Bounded Sybil Attacks)

We prove that V-IC survives when an adversary controls up to $k < \lfloor n/2 \rfloor$ committee seats. The proof structure mirrors Theorem 1: we show $P^{\text{Sybil}}_{\text{match}} > \frac{1}{2}$, then substitute into the IC derivation.

Step 1: Threat model. The adversary controls k validators and casts all k votes in a coordinated, worst-case direction (e.g., all voting "pass" regardless of the ground truth). The remaining $h = n - k > n/2$ honest validators report truthfully with individual accuracy $q_v > \frac{1}{2}$. The majority is determined by all n votes.

Step 2: $P^{\text{Sybil}}_{\text{match}} > \frac{1}{2}$. We condition on the ground truth θ and the adversary's vote direction.

Case A: θ agrees with the adversary's vote direction. The k adversarial votes reinforce the truth, making it easier for the honest majority to prevail. In this case $P^{\text{Sybil}}_{\text{match}} \geq P_{\text{match}}(n, q_v) > \frac{1}{2}$.

Case B: θ opposes the adversary's vote direction. The honest validators must overcome k adversarial votes to form a correct majority. An honest validator v_i matches the overall majority iff at least $\lfloor n/2 \rfloor - k$ of the other $h - 1$ honest peers also vote correctly (since the k adversarial votes contribute to the opposing tally). By the Condorcet jury theorem applied to the h honest validators (with $h > n/2$), the honest bloc retains a strict majority with probability exceeding $\frac{1}{2}$. A truthful v_i agrees with this correct honest majority with probability $\geq q_v^2 + (1 - q_v)^2 > \frac{1}{2}$.

Averaging over both cases (each weighted by $\Pr(\theta = 0)$ and $\Pr(\theta = 1)$) yields $P^{\text{Sybil}}_{\text{match}} > \frac{1}{2}$.

Step 3: IC condition. Since the payoff structure (stake σ, reward $R_{\min}$, shirking baseline $\frac{1}{2}$) is unchanged, the IC derivation of Appendix A.2.4 transfers verbatim with P_{match} replaced by $P^{\text{Sybil}}_{\text{match}}$, yielding condition (13). □

Closed form and Sybil-resistance budget. For the per-task case $m = 1$ and a worst-case adversary that votes uniformly to "pass," the matching probability admits a closed form. Conditional on θ, an honest validator v_i matches the n-vote majority iff at least $\lfloor n/2 \rfloor - k$ of the remaining $h - 1$ honest peers vote with v_i (the k adversarial votes contribute deterministically to the opposing tally). Writing $\alpha_h(q_v) = \sum_{j=\max(0,\lfloor n/2 \rfloor - k)}^{h-1} \binom{h-1}{j} q_v^j (1 - q_v)^{h-1-j}$ for the tail probability that the honest peers reach the threshold conditional on v_i being correct, and $\beta_h(q_v)$ for the corresponding tail conditional on v_i being incorrect, one obtains

$$P^{\text{Sybil}}_{\text{match}}(n, k, q_v) \;=\; q_v\, \alpha_h(q_v) + (1 - q_v)\, \beta_h(q_v).$$

Substituting into (13) defines the *Sybil-resistance budget*:

$$k^*(n, q_v, \tau_{\text{val}}, \sigma, c_u) \;=\; \max\Big\{\, k \in \mathbb{Z}_{\geq 0} \;:\; k < \lfloor n/2 \rfloor \text{ and } (\tau_{\text{val}} + 2\sigma)\big(P^{\text{Sybil}}_{\text{match}}(n, k, q_v) - \tfrac{1}{2}\big) > c_u \,\Big\}.$$

Numerical example. With $n = 9$, $q_v = 0.8$, $\tau_{\text{val}} = \sigma = 1$, and $c_u = 0.5$, the mechanism tolerates up to $k^* = 2$ adversarial seats while maintaining honest-validator IC. Doubling the stake to $\sigma = 2$ raises the budget to $k^* = 3$, confirming that adversary resistance is parameterized by economic cost.

A.4.3 Robustness Details: Correlation and Heterogeneity

This subsection provides the closed-form bounds for relaxations (B) and (C) of §3.1.6.

(B) Residual correlation: closed form of ρ^*. Under the latent-factor mixture model of §3.1.6(B), the matching probability is given by Eq. (14):

$$P^{\rho}_{\text{match}}(n, q_v, \rho) = \rho\left[q_v^2 + (1 - q_v)^2\right] + (1 - \rho)\, P_{\text{match}}(n, q_v).$$

This is a linear function of ρ that is strictly decreasing in ρ whenever $P_{\text{match}}(n, q_v) > q_v^2 + (1 - q_v)^2$, which holds for $n \geq 3$ and $q_v > \frac{1}{2}$ by Proposition 1. Substituting into the IC condition $(\tau_{\text{val}} + 2\sigma)(P^{\rho}_{\text{match}} - \frac{1}{2}) > c_u$ and solving for ρ yields the maximum tolerable correlation:

$$\rho^* \;=\; 1 \;-\; \frac{c_u/(\tau_{\text{val}} + 2\sigma) \;-\; \left[q_v^2 + (1 - q_v)^2 - \frac{1}{2}\right]}{P_{\text{match}}(n, q_v) \;-\; \left[q_v^2 + (1 - q_v)^2\right]}. \tag{16}$$

For $\rho < \rho^*$, V-IC holds strictly. The denominator is positive by Proposition 1, and the numerator is bounded by the IC margin, so $\rho^* \in (0, 1)$ whenever the baseline IC condition is satisfied with slack.

Numerical example. With $n = 9$, $q_v = 0.8$, $\tau_{\text{val}} = \sigma = 1$, and $c_u = 0.5$: $P_{\text{match}}(9, 0.8) \approx 0.984$, $q_v^2 + (1 - q_v)^2 = 0.68$, and $c_u/(\tau_{\text{val}} + 2\sigma) = 0.5/3 \approx 0.167$. Substituting yields $\rho^* \approx 0.72$, tolerating substantial departure from the idealized $\rho = 0$.

Compensation levers. For any fixed $\rho < 1$, the bound ρ^* can be enlarged by (i) raising σ (which increases $\tau_{\text{val}} + 2\sigma$, shrinking the numerator), or (ii) raising n (which increases P_{match}, enlarging the denominator).

(C) Heterogeneous validator quality. With heterogeneous qualities $\{q_{v,i}\}_{i=1}^n$, validator v_i's matching probability $P_{\text{match},i}$ depends on the entire quality profile of the committee. Since each peer votes truthfully and independently with accuracy $q_{v,j}$, the tally of correct votes among the $n-1$ peers, $T = \sum_{j \neq i} \mathbb{1}[r_j = \theta]$, is a sum of independent (but not identically distributed) Bernoulli random variables with mean $\bar{q}_{v,-i} = \frac{1}{n-1} \sum_{j \neq i} q_{v,j}$.

The heterogeneous version of the Condorcet jury theorem (see, e.g., Berend and Paroush, 1998) states that, provided every $q_{v,j} > \frac{1}{2}$, the majority over n truthful votes is correct with probability strictly greater than $\frac{1}{2}$, and the probability tends to 1 as $n \to \infty$ provided $\bar{q}_{v,-i}$ remains bounded above $\frac{1}{2}$.

Combining this with the per-validator agreement probability $q_{v,i}$ yields $P_{\text{match},i} > \frac{1}{2}$ for every i, as long as the quality dispersion is bounded so that no single validator's accuracy is so far below the mean that it cannot benefit from the committee's collective accuracy. The IC condition $(\tau_{\text{val}} + 2\sigma)(P_{\text{match},i} - \frac{1}{2}) > c_u$ then applies to each validator individually, with identical algebraic structure to the homogeneous case once $P_{\text{match},i}$ is computed from the quality profile.

The architectural-heterogeneity requirement of Assumption 1, combined with the producer's incentive to recruit committees of comparable caliber, makes bounded dispersion above $\frac{1}{2}$ the typical operating regime.

In the symmetric special case $q_{v,i} = q_v$ for all i, $P_{\text{match},i}$ collapses to $P_{\text{match}}(n, q_v)$ and Theorem 1 is recovered exactly. □